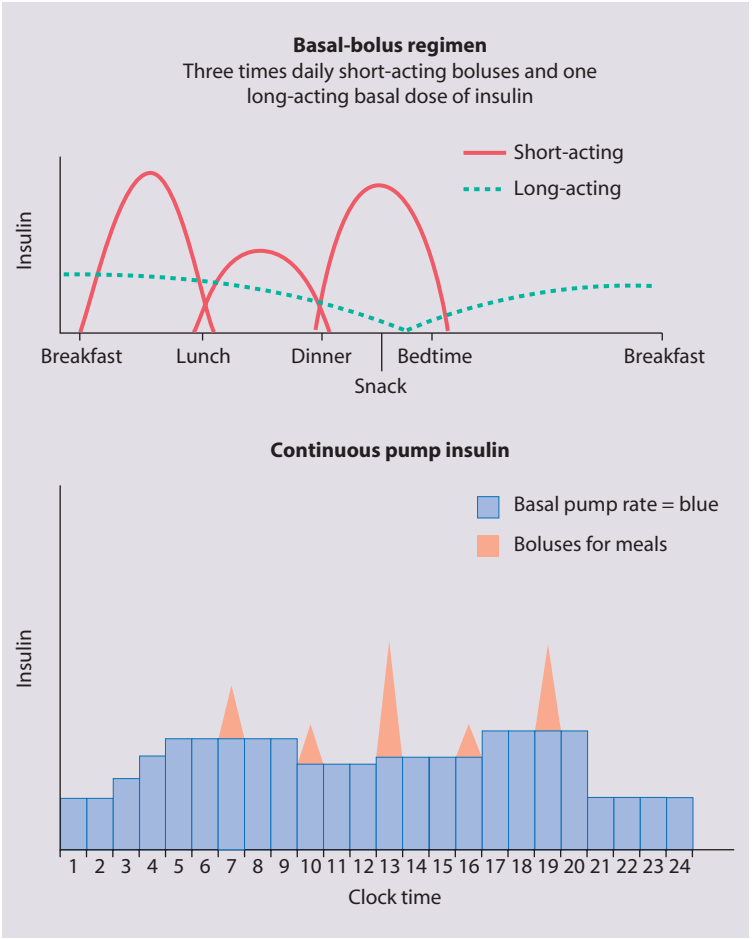


Figure 25.3 Basal-bolus insulin regimen and continuous pump insulin regimen, showing the basal levels of insulin programmed into the pump (blue bars) and the bolus insulin (red pulses) given before each meal/snack according to carbohydrate intake.

of the injection sites is essential to prevent lipohypertrophy or, more rarely, lipoatrophy. The skin should be pinched up and the insulin injected at a 45° angle. Using a long needle or an injection technique that is 'too vertical' causes a painful, bruised intramuscular injection. Shallow intradermal injections can also cause scarring and should be avoided.

Most children are started on an insulin pump or a 3–4 times/day injection regimen ('basal-bolus') with short-acting insulin (e.g. Lispro, Glulisine or Insulin Aspart) being given (bolus) before each meal and snack plus long-acting insulin (e.g. Glargine or Detemir) in the late evening and/or before breakfast to provide insulin background (basal). These treatments both allow greater flexibility by relating the insulin more closely to food intake and exercise (Fig. 25.3). Patients and families are also taught how to correct any sugar above 10 mmol/L between usual meal times by extra short-acting insulin injections. However, the input required by the teams to start these intensive regimens is high, as is the need for a supportive school environment, and some patients and families still rely on twice-daily treatment with premixed insulin.

Shortly after presentation, when some pancreatic function is preserved, insulin requirements often become minimal, the so-called 'honeymoon period'. Requirements subsequently increase to 0.5–1 U/kg or even up to 2 U/kg per day during puberty.

Diet

The diet and insulin regimen need to be matched (Fig. 25.4). The aim is to optimise metabolic control while maintaining normal growth. A healthy diet is recommended, with a high complex carbohydrate and relatively low fat content (<30% of total calories). The diet should be high in fibre, which will provide a sustained release of glucose, rather than refined carbohydrate, which causes rapid swings in glucose levels. 'Carbohydrate counting' allows patients to calculate their likely insulin requirements once their food choice for a meal is known, and taking into account their pre-meal sugar level and post-meal exercise pattern. Learning this balancing act requires a lot of educational input followed by refinement in the light of experience.

Blood glucose monitoring

Regular blood glucose profiles and blood glucose measurements, when a low or high level is suspected, are required to adjust the insulin regimen and learn how changes in lifestyle, food and exercise affect control. A record should be kept in a diary or transferred from the memory of the blood glucose meter. The aim is to maintain blood glucose as near to normal (4–6 mmol/L) as possible. In practice, in order also to avoid hypoglycaemic episodes, this means levels of 4–10 mmol/L in children and 4–8 mmol/L in

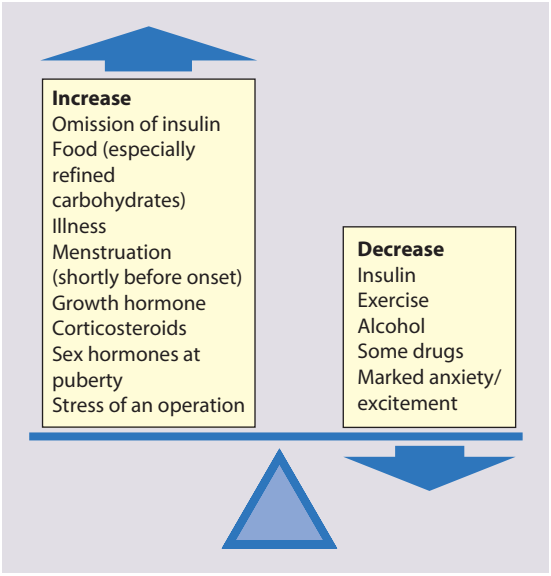


Figure 25.4 Factors affecting blood glucose levels.

adolescents for as much of the time as possible. Realistic goals need to be agreed, with compromises reached about the frequency of monitoring and lifestyle issues, especially in teenagers. During changes in routine (e.g. holidays) or illness, it is not unreasonable to ask for four or more tests per day. In reality, many adolescents test less than once per week, if at all.

Continuous glucose monitoring sensors (CGMS), using subcutaneous or transcutaneous sensors to provide a continuous reading of blood glucose, are now widely available, and refinement is underway to allow these devices to help control the insulin delivered from a pump, or to suggest doses for a given meal composition and size. Continuous glucose monitoring sensors also allow the detection of unexpected asymptomatic episodes of nocturnal hypoglycaemia or times of poor control during the day. Blood ketone testing (often using the same meter as for blood glucose) is mandatory during infections or when control is poor to try to avoid severe ketoacidosis. Urine ketone testing is still used in some centres.

The measurement of glycosylated haemoglobin (HbA_{1c}) is particularly helpful as a guide of overall control over the previous 6–12 weeks and should be checked at least 3 times per year. The level is related to the risk of later complications in a non-linear fashion, such that the risk of complications increases more rapidly with higher levels, but may be misleading if the red blood cell lifespan is reduced, such as in sickle cell trait or if the HbA molecule is abnormal, as in thalassaemia. Since 2009, the units of HbA_{1c} (originally expressed as a % figure) have been changed to an international reporting standard of mmol/mol. Until 2012, results will be reported in both old and new units to give everyone time to become familiar with the new units. A level of ≤58 mmol/mol (7.5%) is seen as an ideal target for patients, but in practice is only achieved in 50% or less of a clinic population in the UK.

Hypoglycaemia

Most children develop well-defined symptoms when their blood glucose falls below about 4 mmol/L. The symptoms are highly individual and change with age, but most complain of hunger, tummy ache, sweatiness, feeling faint or dizzy or of a 'wobbly feeling' in their legs. If unrecognised or untreated, hypoglycaemia may progress to seizures and coma. Parents can often detect hypoglycaemia in young children by their pallor and irritability, sometimes presenting as unreasonable behaviour. If there is any doubt, the blood glucose concentration should be checked or food given.

Treating a 'hypo' at an early stage requires the administration of easily absorbed glucose in the form of glucose tablets (e.g. Lucozade tablets or similar) or a non-diet sugary drink. Children should always have easy access to their hypo remedy, although young children quickly learn to complain of hypo-symptoms in order to leave class or obtain a sweet drink! Oral glucose gels (e.g. Glucogel) are easily and quickly absorbed from the buccal mucosa and so are helpful if the child is unwilling or unable to cooperate to eat. It can be administered by teachers or other helpers. Parents and school should be provided with a glucagon injection kit for the treatment of severe hypoglycaemia, and taught how to administer it intramuscularly to terminate severe hypos. After treatment of 'hypos', parents or carers should give the child some food (usually a biscuit or sandwich) to ensure the blood glucose does not drop again.

Severe hypoglycaemia can usually be predicted (or explained in retrospect – missed meal, heavy exercise). The aim is anticipation and prevention. Hypoglycaemia in an unconscious child brought to hospital is treated with glucose given intravenously.

Diabetic ketoacidosis

Presentation is described in Box 25.2, essential investigations in Box 25.4 and management in Figure 25.5.

Long-term management

The aims of long-term management are:

- Normal growth and development
- Maintaining as normal a home and school life as possible
- Good diabetic control through knowledge and good technique
- Encouraging children to become self-reliant, but with adult supervision until they are able to take responsibility
- Avoidance of hypoglycaemia
- The prevention of long-term complications and an HbA_{1c} of 58 mmol/mol (7.5%) or less.

These aims are difficult to achieve in all patients at all stages of their condition.

Problems in diabetic control

Good blood glucose control is particularly difficult in the following circumstances:

- Eating too many sugary foods, such as sweets taken at odd times, at parties or on the way home from school

Diabetic ketoacidosis

Box 25.4 Essential early investigations

- Blood glucose (>11.1 mmol/L)
- Blood ketones (>3.0 mmol/L)
- Urinary glucose and ketones (both are present)
- Evidence of a precipitating cause, e.g. infection (blood and urine cultures performed)
- Cardiac monitor for T-wave changes of hypokalaemia
- Weight



(b)



(c)

(a) Management priorities

This regimen is initiated if the child is vomiting or has a reduced level of consciousness. Otherwise, even if newly presenting, only subcutaneous insulin is required.

1. Fluids

If in shock, initial resuscitation is with normal saline. Dehydration should then be corrected gradually over 48–72 h (see Fig. 25.5b and c). Rapid rehydration should be avoided as it may lead to cerebral oedema. Initial rehydration fluids need to be taken into account in calculating fluid requirements. Monitor:
- fluid input and output
- electrolytes, creatinine and acid–base status regularly
- neurological state.

Insert central venous line (CVP) and urinary catheter if shocked.
A nasogastric tube is passed for acute gastric dilatation if there is vomiting or depressed consciousness.

2. Insulin

Insulin infusion (0.05–0.1 U/kg per h) is started after 1h, titrating the dose according to the blood glucose. Do not give a bolus. Monitor the blood glucose regularly. Aim for gradual reduction of blood glucose of about 2 mmol/h, as rapid reduction is dangerous. Change to 4% dextrose/ 0.18% saline after 24 h when the blood glucose has fallen to 14 mmol/L to avoid hypoglycaemia.

3. Potassium

Although the initial plasma potassium may be high, it will fall following treatment with insulin and rehydration. Potassium replacement must be instituted as soon as urine is passed. Continuous cardiac monitoring and regular plasma potassium measurements are indicated until the plasma potassium is stable.

4. Acidosis

Although a metabolic acidosis is present, bicarbonate should be avoided unless the child is shocked or not responding to therapy. The acidosis will self-correct with fluid and insulin therapy. Capillary ketones should be monitored.

5. Re-establish oral fluids, subcutaneous insulin and diet

Do not stop the intravenous insulin infusion until 1 h after subcutaneous insulin has been given.

6. Identification and treatment of an underlying cause

Ketoacidosis may be precipitated by an intercurrent infection. Diabetic ketoacidosis causes neutrophilia but not a fever. Antibiotics may be indicated. If the child was known to have diabetes, consider the reason for the ketoacidosis.

Figure 25.5 **(a)** Management priorities in diabetic ketoacidosis. **(b)** Severe dehydration and weight loss from diabetic ketoacidosis. **(c)** Four months later. (Courtesy of Dr Jill Challener.)

- Infrequent or unreliable blood glucose testing. 'Perfect' results are often invented and written down just before clinic to please the diabetes team
- Illness – viral illnesses are common in the young and although it is usually stated that infections cause insulin requirements to increase, in practice the insulin dose required is variable, partly because of reduced food intake. The dose of insulin should be adjusted according to regular blood glucose monitoring. Insulin *must* be continued during times of illness and the urine or blood tested for ketones. If ketosis is increasing along with a rising blood sugar, the family should know how to seek immediate advice to ensure that they increase the soluble insulin dose appropriately or seek medical help for possible intravenous therapy
- Exercise – vigorous or prolonged planned exercise (cross-country running, long-distance hiking, skiing) requires reduction of the insulin dose and increase in dietary intake. Late hypoglycaemia may occur during the night or even the next day, but may be avoided by taking an extra bedtime snack, including slow-acting carbohydrate such as cereal or bread. Less vigorous exercise such as sports lessons in school and spontaneous outdoor play can be managed with an extra snack or a reduction in short-acting insulin before the exercise
- Eating disorders, which are common in young females with diabetes.
- Family disturbance such as divorce or separation
- Inadequate family motivation, support or understanding. As children can never have a 'holiday' from their diabetes, they need a great deal of encouragement to continuously maintain good control. Educational programmes for children and families need to be arranged regularly and matched to their current level of education. Special courses and holiday camps are available; in the UK they are organised by Diabetes UK and local groups.

Management at school

An individualised care plan should be developed by the parents, diabetes team and the school to address the specific needs of the child. This will include the child's dietary needs, requirements to have snacks at specified times and what to do if the child becomes hypoglycaemic or loses consciousness. For younger children, support is needed to help calculate and give the pre-lunch insulin injection or bolus from the pump.

Puberty and adolescence

The rapid growth spurt in early puberty is governed by a complex interaction of hormonal changes, some of which involve insulin and insulin-like growth factors. Growth hormone, oestrogen and testosterone all antagonise insulin action and there is thus an increase in the insulin requirement from the usual 0.5–1.0 U/kg per day of early childhood up to ≥2 U/kg per day. The increase may be especially marked first thing in the morning. The psychological changes accompanying adolescence may make this a time of rebellion where adherence to insulin and dietary regimens is minimal. Diabetic teenagers know that they will not become ill immediately if they cheat with their diet or miss an injection. Some will inevitably test the degree to which the rules can be broken, choosing to ignore the uncomfortable facts of diabetes provided that they 'feel OK'. This usually results in avoidance of blood testing and a tendency to work on the false assumption that feeling well equates with good control. Many teenage girls experiment with crash diets at some time, which are likely to cause major problems in diabetic control. They also learn that glycosuria can be used as an 'aid' to losing weight.

Battles with parents may concentrate on diabetic management instead of the more usual teenage concerns (Table 25.1). Conflict may also extend to involve the professionals of the diabetic team, because of intense anger against the disease which marks them out as different from their peers. Many parents are very protective at this time, whereas teenagers should be encouraged to take responsibility for their diabetes. Health education about smoking, alcohol and contraception may need to be provided. Liaison with a psychologist or child psychiatrist may be helpful. The professionals of the diabetic team may need to encourage diabetic teenagers to take better care of themselves. It is usually unhelpful to give lectures about the long-term risks to health, as these are likely to be seen as irrelevant by the teenagers. However, they may be helped if:

- There are clear short-term goals agreed by the patient
- Their efforts to improve their diabetic control, e.g. an improving or satisfactory HbA_{1c} level, are communicated promptly and enthusiastically
- There is a united team approach, with agreement between professionals of the essentials they wish to promote and clear, unambiguous guidelines for health and diabetic management
- Peer group pressure is used to promote health. Activities such as holidays, etc., that allow teenagers to participate while learning about their diabetic management are encouraged. They may also benefit by being used as teachers of younger children.

Successful long-term diabetic management depends on education and increasing self-reliance and responsibility.

After many years in a children's clinic, it can be difficult for the patient and family to move to the adult care environment. This transition is helped by discussing and planning the move well ahead of the time, and by the provision of joint clinics with the adult diabetologists through to the early twenties or end of tertiary education. Special peri-conceptional clinics have been established in some centres to help achieve near-ideal control before conception in planned pregnancies. Conception of a fetus with a high HbA_{1c} increases the risks of congenital abnormalities in the offspring.

Table 25.1 How diabetes interferes with normal adolescence

Normal adolescence	How diabetes interferes
Physical and sexual maturation	Delayed sexual maturation
	Invasion of privacy with frequent medical examinations
Conformity with peer group	Meals must be eaten on time
	Frequent injections and blood tests
Self-image	Hypoglycaemic attacks show that they are different
Self-esteem	Impaired body image
Independence from parents	Parental over-protection and reluctance to allow their child to be away from home
	Battles over diabetes
Economic independence	Loading of insurance premiums
	Discrimination by employers
	Statutory rules against becoming a pilot or driving heavy goods or public service vehicles

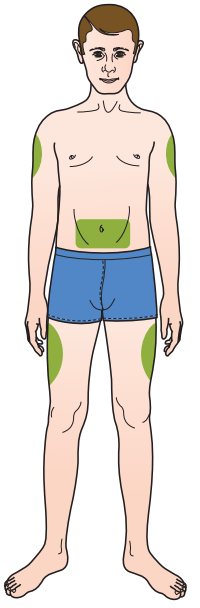
Summary

Regular assessment of the child with diabetes

Assessment of diabetic control:
- Any episodes of hypoglycaemia, diabetic ketoacidosis, hospital admission?
- Is there still awareness of hypoglycaemia?
- Absence from school? School supportive of diabetes care?
- Interference with normal life?
- HbA_{1c} results – 58 mmol/mol (7.5%) or less?
- Diary of blood glucose results – if monitoring, is he reacting to results?
- Insulin regimen – appropriate? Correction bolus doses given?
- Lipohypertrophy or lipoatrophy (Fig. 25.6a and b) at injection sites?
- Diet – healthy diet, manipulating food intake and insulin to maintain good control?

General overview (periodic):
- Normal growth and pubertal development, avoiding obesity – measure each visit
- Blood pressure check for hypertension yearly (age-specific centiles)
- Renal disease – screening for microalbuminuria yearly from 12 years
- Eyes – photography for retinopathy or cataracts, yearly from 12 years
- Feet – maintaining good care – yearly
- Screening for coeliac and thyroid disease at diagnosis, thyroid screening yearly, coeliac again after 3 years or if weight gain poor.
- Annual reminder to have flu vaccination

Knowledge and psychosocial aspects:
- Good understanding of diabetes, would participation/holidays with other diabetic children be beneficial? Member of Diabetes UK?
- Becoming self-reliant, but appropriate supervision at home, school, diabetic team?
- Taking exercise, sport? Diabetes not interfering with it?
- Leading as normal life as possible?
- Smoking, alcohol?
- Is 'hypo' treatment readily available? Is stepped approach known?
- What are the main issues for the patient? Are there short-term goals to allow engagement with improving control?



(a)

Injection sites – check for lipohypertrophy or lipoatrophy



(b)

Figure 25.6 Lipohypertrophy from insulin injections. (a) Injection sites. (b) Lipohypertrophy (arrow) from insulin injections.

Prevention of long-term complications

It has been shown that meticulous diabetic control delays or prevents diabetic retinopathy and nephropathy and, if retinopathy occurs, it can slow the progression. There is also evidence that good early control reduces the risk of later complications, even if control deteriorates later in life. Levels of glycosylated haemoglobin above 58 mmol/mol (7.5%) are related to the risk of later complications in an almost exponential fashion, and so the ideal is to aim for below this level as much as possible.

Although long-term health problems are uncommon during childhood, there needs to be regular review for long-term complications and associated illnesses:

- *Growth and pubertal development*. Some delay in the onset of puberty may occur. Obesity is common, especially in females, if their insulin dose is not reduced towards the end of puberty. Intensive insulin regimens increase the risk of excessive weight gain and BMI should be plotted at each clinic visit
- *Blood pressure* – must be checked at least once a year for evidence of hypertension
- *Renal disease* – the detection of microalbuminuria is an early sign of nephropathy and should be screened annually in teenagers
- *Eyes* – retinopathy or cataracts requiring treatment are rare in children but should be monitored annually after 5 years of diabetes or from the onset of puberty, ideally with retinal photography
- *Feet* – children should be encouraged to take good care of their feet from an early age, to avoid tight shoes and treat any infections early
- *Other associated illnesses* – coeliac disease and thyroid disease are more common in type 1 diabetes and easily missed clinically, so screening for them is recommended at diagnosis and subsequently (thyroid function yearly post-diagnosis and coeliac screening by tissue transglutaminase levels after 3 years) or if suspected clinically. There should be a low threshold for investigating for other autoimmune disorders (rheumatoid, vitiligo, etc.).

Hypoglycaemia

Hypoglycaemia is a common problem in neonates during the first few days of life (see Chapter 10). Thereafter, it is uncommon in non-diabetics. It is often defined as a plasma glucose <2.6 mmol/L, although the development of clinical features will depend on whether other energy substrates can be utilised. Clinical features include:

- Sweating
- Pallor
- Central nervous system signs of irritability, headache, seizures and coma.

The neurological sequelae may be permanent if hypoglycaemia persists and include epilepsy, severe learning difficulties and microcephaly. This risk is greatest in early childhood during the period of most rapid brain growth.

Infants have high energy requirements and relatively poor reserves of glucose from gluconeogenesis and glycogenesis. They are at risk of hypoglycaemia with fasting. Infants should never be starved for more than 4 h, e.g. preoperatively. A blood glucose should be checked in any child who:

- becomes septicaemic or appears seriously ill. **ABC** then **DEFG** – '**D**on't **E**ver **F**orget **G**lucose'
- has a prolonged seizure
- develops an altered state of consciousness.

This is often done at the bedside, using glucose-sensitive strips, the accuracy of which is improved by use of a meter. However, the strips only indicate that the glucose is within a low range of values and any low reading must always be confirmed by laboratory measurement.

If the cause of the hypoglycaemia is unknown, *it is vital that blood is collected at the time of the hypoglycaemia* and the first available urine sent for analysis, so that a valuable opportunity for making the diagnosis is not missed (Box 25.5).

Box 25.5 Tests to perform when hypoglycaemia is present

Blood

- Confirm hypoglycaemia with laboratory blood glucose
- Growth hormone, IGF-1, cortisol, insulin, C-peptide, fatty acids, ketones (acetoacetate, 3-hydroxybutyrate), glycerol, branched-chain amino acids, acylcarnitine profile, lactate, pyruvate

First urine after hypoglycaemia

- Organic acids
- Consider saving blood and urine for toxicology, e.g. salicylate, sulphonylurea

Causes

These are listed in Box 25.6.

Ketotic hypoglycaemia is a poorly-defined entity in which young children readily become hypoglycaemic following a short period of starvation, probably due to limited reserves for gluconeogenesis. The child is often short and thin and the insulin levels are low. Regular snacks and extra glucose drinks when ill will usually prevent hypoglycaemia. The condition resolves spontaneously in later life. A number of rare endocrine and metabolic disorders may present with hypoglycaemia at almost any age in childhood. Hepatomegaly would suggest the possibility of an inherited glycogen storage disorder, in which hypoglycaemia can be profound.

Transient neonatal hypoglycaemia in neonates may be due to exposure to high levels of insulin in utero if mothers are diabetic or glucose intolerant. In contrast, recurrent, severe neonatal hypoglycaemia may be

Box 25.6 Causes of hypoglycaemia beyond the immediate neonatal period

Fasting

- *Insulin excess*
  - Excess exogenous insulin, e.g. in diabetes mellitus/insulin given surreptitiously
  - β-cell tumours/disorders – persistent hypoglycaemic hyperinsulinism of infancy (PHHI, previously called nesidioblastosis), insulinoma
  - Drug-induced (sulphonylurea)
  - Autoimmune (insulin receptor antibodies)
  - Beckwith syndrome
- *Without hyperinsulinaemia*
  - Liver disease
  - Ketotic hypoglycaemia of childhood
  - Inborn errors of metabolism, e.g. glycogen storage disorders
  - Hormonal deficiency: GH↓, ACTH↓, Addison disease, congenital adrenal hyperplasia

Reactive/non-fasting

- Galactosaemia
- Leucine sensitivity
- Fructose intolerance
- Maternal diabetes
- Hormonal deficiency
- Aspirin/alcohol poisoning.

caused by persistent hypoglycaemic hyperinsulinism of infancy (PHHI, formerly called 'nesidioblastosis'). This is a rare disorder of infancy where there are gene mutations of various pathways leading to dysregulation of insulin release by the islet cells of the pancreas leading to profound non-ketotic hypoglycaemia. Treatment with high-concentration dextrose solutions and diazoxide (plus other medications) may be required to maintain safe blood sugar levels pending investigation. Special scans reveal that up to 40% of cases are caused by localised lesions in the pancreas amenable to partial resection, although the majority of cases either require long-term medication or total pancreatectomy with the attendant risk of diabetes and exocrine pancreatic insufficiency.

Treatment

Hypoglycaemia can usually be corrected with an intravenous infusion of glucose (2 ml/kg of 10% dextrose followed by 10% dextrose infusion). Care must be taken to avoid giving an excess volume as the solution is hypertonic and could cause cerebral oedema. If there is delay in establishing an infusion or failure to respond, glucagon is given intramuscularly (0.5–1 mg). If a higher concentration than a 10% solution is required in a neonate, the low sugar is highly likely to be secondary to hyperinsulinism.

Corticosteroids may also be used if there is a possibility of hypopituitarism or hypoadrenalism. The correction of hypoglycaemia must always be documented with satisfactory laboratory glucose measurements.

Summary

Hypoglycaemia

- Should be excluded in any child with septicaemia, who is seriously ill, has a prolonged seizure or altered state of consciousness ('**D**on't **E**ver **F**orget **G**lucose')
- Low blood glucose on bedside testing must be confirmed by laboratory measurement
- If the cause is unknown, diagnostic blood and urine samples should, if possible, be taken at the time.

Hypothyroidism

There is only a small amount of thyroxine transfer from the mother to the fetus, although severe maternal hypothyroidism can affect the developing brain. The fetal thyroid predominantly produces 'reverse T_3', a derivative of T_3 which is largely inactive. After birth, there is a surge in the level of thyroid-stimulating hormone (TSH) which is accompanied by a marked rise in T_4 and T_3 levels. The TSH declines to the normal adult range within a week. Preterm infants may have very low levels of T_4 for the first few weeks of life, while the TSH is within the normal range; under these circumstances, additional thyroxine is not required.

Congenital hypothyroidism

Detection of congenital hypothyroidism is important, as it is:

- Relatively common, occurring in 1 in 4000 births
- One of the few preventable causes of severe learning difficulties.

Causes of congenital hypothyroidism are:

- *Maldescent of the thyroid and athyrosis* – the commonest cause of sporadic congenital hypothyroidism. In early fetal life, the thyroid migrates from a position at the base of the tongue (sublingual) to its normal site below the larynx. The thyroid may fail to develop completely or partially. In maldescent, the thyroid remains as a lingual mass or a unilobular small gland. The reason for this failure of formation or migration is not well understood
- *Dyshormonogenesis*, an inborn error of thyroid hormone synthesis, in about 5–10% of cases, although commoner in some ethnic groups with consanguineous marriage
- *Iodine deficiency*, the commonest cause of congenital hypothyroidism worldwide but rare in the UK. It can be prevented by iodination of salt in the diet

Box 25.7 Clinical features of hypothyroidism

Congenital	Acquired
Usually asymptomatic and picked up on screening. Otherwise:	Females > males
Failure to thrive	Short stature/growth failure
Feeding problems	Cold intolerance
Prolonged jaundice	Dry skin
Constipation	Cold peripheries
Pale, cold, mottled dry skin	Bradycardia
Coarse facies	Thin, dry hair
Large tongue	Pale, puffy eyes with loss of eyebrows
Hoarse cry	Goitre
Goitre (occasionally)	Slow-relaxing reflexes
Umbilical hernia	Constipation
Delayed development	Delayed puberty
	Obesity
	Slipped upper femoral epiphysis
	Deterioration in school work
	Learning difficulties



Figure 25.7 Untreated congenital hypothyroidism.

- Hypothyroidism due to *TSH deficiency* – isolated TSH deficiency is rare (<1% of cases) and is usually associated with panhypopituitarism, which usually manifests with growth hormone, gonadotrophin and ACTH deficiency leading to hypoglycaemia or micropenis and undescended testes in affected boys before the hypothyroidism becomes evident.

The clinical features (Box 25.7 and Fig. 25.7) are difficult to differentiate from normal in the first month of life, but become more prominent with age. There is a slight excess of other congenital abnormalities, especially heart defects.

Most infants with congenital hypothyroidism are detected on routine neonatal biochemical screening (Guthrie test), performed on all newborn infants, by identifying a raised TSH in the blood. However, thyroid dysfunction secondary to pituitary abnormalities may not be picked up at neonatal screening as they will have a low TSH. In some countries T_4 is also measured. Treatment with thyroxine is started at 2–3 weeks of age.

Early treatment of congenital hypothyroidism is essential to prevent learning difficulties. With neonatal screening, the results of long-term intellectual development have been satisfactory and intelligence should be in the normal range for the majority of children. Treatment is lifelong with oral replacement of thyroxine, titrating the dose to maintain normal growth, TSH and T_4 levels.

Juvenile hypothyroidism

This is usually caused by autoimmune thyroiditis. There is an increased risk in children with Down or Turner syndrome and of developing other autoimmune disorders, e.g. vitiligo, rheumatoid arthritis, diabetes mellitus. In some families, Addison disease may also occur.

The clinical features are listed in Box 25.7. It is commoner in females. There is growth failure accompanied by delayed bone age. Goitre is often present but this may also be physiological in pubertal girls. Treatment is with thyroxine.

Summary

Congenital hypothyroidism

- Is identified on routine neonatal biochemical screening (Guthrie test)
- Although present antenatally, treatment started soon after birth results in satisfactory intellectual development.

Hyperthyroidism

This usually results from Graves disease (autoimmune thyroiditis), secondary to the production of thyroid-stimulating immunoglobulins (TSIs). The clinical

Box 25.8 Clinical features of hyperthyroidism

Systemic	Eye signs (uncommon in children)
Anxiety, restlessness	Exophthalmos
Increased appetite	Ophthalmoplegia
Sweating	Lid retraction
Diarrhoea	Lid lag
Weight loss	
Rapid growth in height	
Advanced bone maturity	
Tremor	
Tachycardia, wide pulse pressure	
Warm, vasodilated peripheries	
Goitre (bruit)	
Learning difficulties/behaviour problems	
Psychosis	

features are similar to those in adults, although eye signs are less common (Box 25.8 and Fig. 25.8). It is most often seen in teenage girls. The levels of thyroxine (T_4) and/or tri-iodothyronine (T_3) are elevated and TSH levels are suppressed to very low levels. Antithyroid peroxisomal antibodies may also be present which may eventually result in spontaneous resolution of the thyrotoxicosis but subsequently cause hypothyroidism (so-called hashitoxicosis).

The first-line of treatment is medical, with drugs such as carbimazole or propylthiouracil that interfere with thyroid hormone synthesis. Initially, β-blockers can be added for symptomatic relief of anxiety, tremor and tachycardia. There is a risk of neutropenia from anti-thyroid medication and all families should be warned to seek urgent help and a blood count if sore throat and high fever occur on starting treatment. Medical treatment is given for about 2 years, which should control the thyrotoxicosis, but the eye signs may not resolve. When medical treatment is stopped, 40–75% relapse. A second course of drugs may then be given or surgery in the form of subtotal thyroidectomy will usually result in permanent remission. Radioiodine treatment is simple and is no longer considered to result in later neoplasia. Follow-up is always required as thyroxine replacement is often needed for subsequent hypothyroidism.

Neonatal hyperthyroidism may occur in infants of mothers with Graves disease from the transplacental transfer of TSIs. Treatment is required as it is potentially fatal, but it resolves spontaneously with time.



Figure 25.8 Exophthalmos in Graves disease.

Parathyroid disorders

Parathyroid hormone (PTH) promotes bone formation via bone-forming cells (osteoblasts). However, when calcium levels are low, PTH promotes bone resorption via osteoclasts, increases renal uptake of calcium and activates metabolism of vitamin D to promote gut absorption of calcium. In hypoparathyroidism, which is rare in childhood, in addition to a low serum calcium, there is a raised serum phosphate and a normal alkaline phosphatase. The parathyroid hormone level is very low. Severe hypocalcaemia leads to muscle spasm, fits, stridor and diarrhoea. It is a common problem in premature infants, and increasingly seen as a presentation of severe rickets (see Ch. 12). Other causes are rare in childhood.

Hypoparathyroidism in infants is usually due to a congenital deficiency (DiGeorge syndrome), associated with thymic aplasia, defective immunity, cardiac defects and facial abnormalities. In older children, hypoparathyroidism is usually an autoimmune disorder associated with Addison disease.

In *pseudohypoparathyroidism* there is end-organ resistance to the action of parathyroid hormone caused by a mutation in a signalling molecule. Serum calcium and phosphate levels are abnormal but the parathyroid hormone levels are normal or high. Other abnormalities are short stature, obesity, subcutaneous nodules, short fourth metacarpals and learning difficulties. There may be teeth enamel hypoplasia and calcification of the basal ganglia. A related state, in which there are the physical characteristics of pseudohypoparathyroidism but the calcium, phosphate and PTH are all normal, is called *pseudopseudohypoparathyroidism*. There may be a positive family history of both disorders in the same kindred.

Treatment of acute symptomatic hypocalcaemia is with an intravenous infusion of calcium gluconate. The 10% solution of calcium gluconate must be diluted as extravasation of the infusion will result in severe skin damage. Chronic hypocalcaemia is treated with oral calcium and high doses of vitamin D analogues, adjusting the dose to maintain the plasma calcium concentration just below the normal range. Hypercalcuria is to be avoided as it may cause nephrocalcinosis and so the urinary calcium excretion should be monitored.

Hyperparathyroidism results in a high calcium level, which in turn produces constipation, anorexia, lethargy and behavioural effects, polyuria and polydipsia. Bony erosions of the phalanges may be seen on a wrist radiograph. In neonates and young children, it is associated with some rare genetic abnormalities (e.g. William syndrome), but in later childhood can be secondary to adenomas occurring spontaneously or as part of the multiple endocrine neoplasia (MEN) syndromes. Severe hypercalcaemia is treated with rehydration, diuretics and bisphosphonates.

Adrenal cortical insufficiency

Congenital adrenal hyperplasia is the commonest non-iatrogenic cause of insufficient cortisol and mineralocorticoid secretion (see Ch. 11).

Primary adrenal cortical insufficiency (Addison disease) is rare in children. It may result from:

- An autoimmune process, sometimes in association with other autoimmune endocrine disorders, e.g. diabetes mellitus, hypothyroidism, hypoparathyroidism
- Haemorrhage/infarction – neonatal, meningococcal septicaemia (usually fatal)
- X-linked adrenoleucodystrophy, a rare neurodegenerative metabolic disorder
- Tuberculosis, now rare.

Adrenal insufficiency may also be secondary to hypopituitarism from hypothalamic–pituitary disease or from hypothalamic–pituitary–adrenal suppression following long-term corticosteroid therapy.

Presentation

Infants present acutely (Box 25.9) with a salt-losing crisis, hypotension and/or hypoglycaemia. Dehydration may follow a gastroenteritis-like illness, from which the child recovers until the next episode. In older children, presentation is usually with chronic ill health and pigmentation (Fig. 25.9).

Diagnosis

This is made by finding hyponatraemia and hyperkalaemia, often associated with a metabolic acidosis and hypoglycaemia. The plasma cortisol is low and the plasma ACTH concentration high (except in hypopituitarism). With an ACTH (Synacthen) test, plasma cortisol concentrations remain low in both primary adrenal failure and in long-standing pituitary/hypothalamic Addison disease. A normal response excludes adrenal cortical insufficiency.

Management

An adrenal crisis requires urgent treatment with intravenous saline, glucose and hydrocortisone. Long-term treatment is with glucocorticoid and mineralocorticoid replacement. The dose of glucocorticoid needs to be increased by three times at times of illness or for an operation. Parents are taught how to inject intramuscular hydrocortisone in an emergency. All children at risk of an adrenal crisis should wear a MedicAlert bracelet.

Box 25.9 Features of adrenal cortical insufficiency

Acute	Chronic
Hyponatraemia	Vomiting
Hyperkalaemia	Lethargy
Hypoglycaemia	Brown pigmentation (gums, scars, skin creases)
Dehydration	Growth failure
Hypotension	
Circulatory collapse	



Figure 25.9 Buccal pigmentation in adrenal cortical insufficiency (Addison disease). This 9-year-old boy presented with salt craving and pigmentation. (Courtesy of Dr Steven Robinson.)

Summary

Adrenal cortical insufficiency

- Usually due to corticosteroid therapy, congenital adrenal hyperplasia or, rarely, Addison disease
- May result in an adrenal crisis requiring urgent treatment.

Cushing syndrome

Glucocorticoid excess in children is usually a side-effect of long-term glucocorticoid treatment (intravenous, oral or, more rarely, inhaled, nasal or topical) for conditions such as the nephrotic syndrome, asthma or, in the past, for severe bronchopulmonary dysplasia (Box 25.10 and Fig. 25.10). Corticosteroids are potent growth suppressors and prolonged use in high dosage will lead to reduced adult height and osteopenia. This unwanted side-effect of systemic corticosteroids is markedly reduced by taking corticosteroid medication in the morning on alternate days.

Other causes of glucocorticoid excess are rare. It may be ACTH-driven, from a pituitary adenoma, usually in older children, or from ectopic ACTH-producing tumours, but these almost never occur in children. ACTH-independent disease is usually from corticosteroid therapy, but may be from adrenocortical tumours

Box 25.10 Clinical features of Cushing syndrome

- Growth failure/short stature
- Face and trunk obesity
- Red cheeks
- Hirsutism
- Striae
- Hypertension
- Bruising
- Carbohydrate intolerance
- Muscle wasting and weakness
- Osteopenia
- Psychological problems.

(benign or malignant), when there may also be virilisation; these usually occur in young children. A diagnosis of Cushing syndrome is often questioned in obese children. Most obese children from dietary excess are of above-average height, in contrast to children with Cushing syndrome, who are short and have growth failure.

If Cushing syndrome is a possibility, then the normal diurnal variation of cortisol (high in the morning, low at midnight) may be shown to be lost – in Cushing syndrome the midnight concentration is also high. The 24-h urine free cortisol is also high. After the administration of dexamethasone, there is failure to suppress the plasma 09.00 h cortisol levels. Adrenal tumours are identified on CT or MRI scan of the abdomen and a pituitary adenoma on MRI brain scan. Adrenal tumours are usually unilateral and are treated by adrenalectomy and radiotherapy if indicated. Pituitary adenomas are best treated by trans-sphenoidal resection, but radiotherapy can be used.

Inborn errors of metabolism

Although individually rare, inborn errors of metabolism are an important cause of paediatric morbidity and mortality. The specialised nature of the diagnostic tests and subsequent management often means that these patients are managed in specialist centres. However, as the prognosis for most patients depends upon the speed of diagnosis, all doctors need to be familiar with their variable presentation and diagnosis. It is often assumed that a precise knowledge of a large number of biochemical pathways is necessary to make a diagnosis, but in fact a more than adequate diagnostic approach can be based on the correct use of only a few screening tests.

Presentation

An inborn error of metabolism may be suspected before birth from a positive family history or previous unexplained deaths in the family.

After birth, inborn errors of metabolism usually, but not invariably, present in one of five ways:

- As a result of newborn screening, e.g. phenylketonuria (PKU), or family screening, e.g. familial hypercholesterolaemia

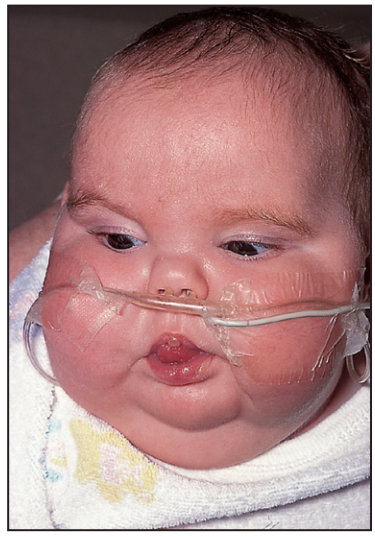


Figure 25.10 Facial obesity from prolonged course of high-dose corticosteroids for bronchopulmonary dysplasia in a preterm infant. Such prolonged courses are no longer used. Additional oxygen therapy is being given via nasal cannulae.

- After a short period of apparent normality, with a severe neonatal illness with poor feeding, vomiting, encephalopathy, acidosis, coma and death, e.g. organic acid or urea cycle disorders
- As an infant or older child with an illness similar to that described above but with hypoglycaemia as a prominent feature or as an ALTE (acute life-threatening episode) or near-miss 'cot death', e.g. a fat oxidation defect such as medium-chain acyl-CoA dehydrogenase deficiency (MCADD)
- In a subacute way, after a period of normal development, with regression, organomegaly and coarse facies, e.g. mucopolysaccharide disease or other lysosomal storage disorder or with enlargement of the liver and/or spleen alone, with or without accompanying biochemical upset such as hypoglycaemia, e.g. glycogen storage disease
- As a dysmorphic syndrome.

Newborn screening

The parents of all babies born in the UK are offered a screening test to detect hypothyroidism and phenylketonuria (PKU). The tests are done on a spot of blood from a heel-prick collected onto a filter paper. Although it is technically possible to screen for a much larger group of disorders, this has been resisted in the UK. However, the screening programme has been extended to include cystic fibrosis, haemoglobinopathies and the metabolic disorder MCADD.

Amino acid disorders

Phenylketonuria

This occurs in 1 in 10 000–15 000 live births in the UK. It is either due to a deficiency of the enzyme phenylalanine hydroxylase (classical PKU) or in the

synthesis or recycling of the biopterin cofactor for this enzyme. Untreated, it usually presents with developmental delay at 6–12 months of age. There may be a musty odour due to the metabolite phenylacetic acid. Many affected children are fair-haired and blue-eyed and some develop eczema and seizures. Fortunately, most affected children are detected through the national biochemical screening programme (Guthrie test).

Treatment of classical PKU is with restriction of dietary phenylalanine, while ensuring there is sufficient for optimal physical and neurological growth. The blood plasma phenylalanine is monitored regularly. The current recommendation is to maintain the diet throughout life. This is particularly important during pregnancy, when high maternal phenylalanine levels may damage the fetus.

Cofactor defects, which have a much poorer prognosis than classical PKU, are treated with a diet low in phenylalanine and neurotransmitter precursors.

Homocystinuria

This is due to cystathionine synthetase deficiency. Presentation is with developmental delay and eventually subluxation of the ocular lens (ectopia lentis). There is progressive learning difficulty, psychiatric disorders and convulsions. Skeletal manifestations resemble Marfan syndrome. The complexion is usually fair with brittle hair. Thromboembolic episodes may occur at any age. Almost half respond to large doses of the coenzyme pyridoxine. Those who do not respond are treated with a low-methionine diet supplemented with cysteine and with the addition of the re-methylating agent betaine.

Tyrosinaemia

Tyrosinaemia (type 1) is a rare autosomal recessive disorder caused by a deficiency of fumarylacetoacetase. Accumulation of toxic metabolites results in damage to the liver (leading to liver failure) and renal tubules (resulting in Fanconi syndrome). Untreated the disorder is fatal, but effective therapy is now available with a drug called NTBC, which inhibits an enzyme required in the catabolism of tyrosine, together with a diet low in tyrosine and phenylalanine.

Disorders presenting acutely in the neonatal period

This group includes:

- Disorders of the catabolic pathways of several essential amino acids (the branched-chain amino acids, leucine, isoleucine and valine, and odd-chain amino acids, e.g. threonine) to cause maple syrup urine disease and other organic acid disorders
- Defects in the urea cycle
- A disorder of carbohydrate metabolism – classical galactosaemia.

In these disorders, the affected child is normal at birth and after several days develops non-specific signs and symptoms shared with other more common neonatal disorders, such as generalised infection. The most common patterns of illness are:

- Vomiting, acidosis and circulatory disturbance, followed by depressed consciousness and convulsions – suggestive of one of the organic acidaemias
- Neurological features of lethargy, refusal to feed, hypotonia, drowsiness, unconsciousness and apnoea – suggestive of primary defects of the urea cycle. Improvement when given intravenous fluids but relapse if milk feeds are restarted is characteristic of classical galactosaemia.

Diagnosis is with a 'metabolic screen' in addition to the standard investigations for unwell infants. The 'metabolic screen' varies between laboratories and should be discussed with the specialist laboratory before collecting samples. The urgency should also be indicated. Both blood and urine samples are likely to be required. A simple bedside test for ketones can be helpful as heavy ketosis and acidosis in an encephalopathic infant is strongly suggestive of an organic acid disorder. In patients with acidosis, calculation of the anion gap (the sum of serum concentrations of sodium and potassium minus the sum of the concentrations of chloride and bicarbonate) can be helpful. Values >25 mmol/L (normal 12–16 mmol/L) are usually secondary to an organic acidaemia. It is good practice to collect all urine passed by the infant for possible future analysis (or until a diagnosis is established), as well as collecting a sample of blood before any blood transfusion in case the latter interferes with the interpretation of laboratory tests. Both short-term and long-term management depend on the underlying diagnosis. In the immediate emergency situation, removal of toxic metabolites and limitation of catabolism have the highest priority. Transfer to a neonatal intensive care unit, mechanical ventilation and haemodialysis are often required. Long-term management involves skilled dietetic support as well as the use of specific medications depending on the underlying diagnosis.

Disorders of carbohydrate metabolism

Galactosaemia

This rare, recessively inherited disorder results from deficiency of the enzyme galactose-1-phosphate uridyltransferase, which is essential for galactose metabolism. When lactose-containing milk feeds such as breast or infant formula are introduced, affected infants feed poorly, vomit and develop jaundice and hepatomegaly and hepatic failure (see Ch. 20). Chronic liver disease, cataracts and developmental delay are inevitable if the condition is untreated. Management is with a lactose- and galactose-free diet for life. Even if treated early, there are usually moderate learning difficulties (adult IQ 60–80).

Glycogen storage disorders

These mostly recessively inherited disorders have specific enzyme defects which prevent mobilisation of glucose from glycogen, resulting in an abnormal

Table 25.2 Some of the glycogen storage disorders

Type	Enzyme defect	Onset	Liver	Muscle	Comments
Type I (von Gierke)	Glucose-6-phosphatase	Infant	+++	–	See Figure 25.11 Enlarged liver and kidneys Growth failure. Hypoglycaemia Good prognosis
Type II (Pompe)	Lysosomal α-glucosidase	Infant	++	+++	Hypotonia and cardiomegaly at several months. Enzyme replacement therapy (Myozyme). Death from heart failure
Type III (Cori)	Amylo-1,6-glucosidase	Infant	++	+	Milder features of type I, but muscles may be affected Good prognosis
Type V (McArdle)	Phosphorylase	Child	–	++	Temporary weakness and cramps muscles after exercise Myoglobinuria in later life

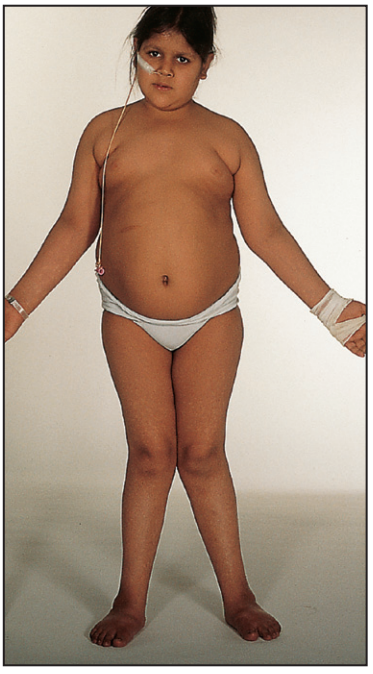


Figure 25.11 Type I glycogen storage disease in a 12-year-old girl. There is truncal obesity with a distended abdomen from an enlarged liver; short stature and hypotrophic muscles; 'doll' facies; nasogastric feeding to maintain blood glucose levels overnight.

storage of glycogen in liver and/or muscle. There are nine main enzyme defects, some of which are shown in Table 25.2. The disorder may predominantly affect muscle (e.g. types II, V), leading to skeletal muscle weakness. In type II (Pompe disease) there is generalised intralysosomal storage of glycogen. The heart is severely affected, leading to death from cardiomyopathy. In other types (e.g. I, III) the liver is the main organ of storage, and hepatomegaly and hypoglycaemia are prominent (Fig. 25.11). Long-term complications of type I include hyperlipidaemia, hyperuricaemia, the development of hepatic adenomas and cardiovascular disease.

Management is to maintain blood glucose by frequent feeds or by carbohydrate infusion via a gastrostomy or nasogastric tube in infancy. In older children, glucose levels can be maintained using slow-release oligosaccharides (corn starch). In type II (Pompe disease), treatment with enzyme replacement therapy (Myozyme) is now available. In type III disorder, a high-protein diet is recommended to prevent growth retardation and myopathy.

Hyperlipidaemia

Hyperlipidaemia is one of the main risk factors for coronary heart disease. Identification and treatment of hyperlipidaemia in childhood may delay the onset of cardiovascular disease in later life.

Children should be screened for hyperlipidaemia if they are at increased risk – if a parent or grandparent has a history of coronary heart disease before 55 years of age or if there is a family history of a lipid disorder. At present, screening all children is not thought justifiable in view of the many uncertainties about selecting who should be treated, what treatment should be given and its effect on outcome.

If the serum cholesterol is high (>5.3 mmol/L) on random testing, fasting serum cholesterol, triglyceride and low-density lipoprotein (LDL) and high-density lipoprotein (HDL) cholesterol are measured. Secondary

causes of hypercholesterolaemia should be considered, such as obesity, hypothyroidism, diabetes mellitus, nephrotic syndrome and obstructive jaundice.

Familial hypercholesterolaemia (FH)

This autosomal dominant disorder of lipoprotein metabolism is due to a defect in the LDL receptor. About 1 in 500 of the population are affected. The serum LDL cholesterol concentration is markedly raised (>3.3 mmol/L). The condition is associated with premature coronary heart disease, which occurs in half by 50 years of age in males and by 60 years in females. Skin and tendon xanthomata (Fig. 25.12) may be present, but are uncommon in childhood. Drug therapy is considered in children aged 10 years and older and depends on how high the LDL cholesterol concentration is raised, if there is a family history of premature coronary heart disease (<55 years of age), if there is evidence of tissue lipid deposition (xanthomata or bruits) and other non-lipid risk factors, e.g. diabetes. The main drugs used are the non-systemically acting bile acid sequestrants and more recently the HMG-CoA reductase inhibitors, the statins. Although bile acid sequestrants are moderately effective, compliance remains a major problem with them. Statins have been shown to be effective in children, without adverse effects on growth, maturation or endocrine function. The fibrate drug fenofibrate has also been shown to reduce LDL cholesterol and to be well tolerated by children and adolescents.



Figure 25.12 Severe skin xanthomata. In this child, it was secondary to liver failure and resolved within weeks of liver transplantation.

Homozygous disease is very rare and much more severe, causing xanthomata in childhood and clinical cardiovascular disease in the second decade. Affected children require referral to a specialist centre. Response to drugs is variable, depending on the gene mutation. Liver transplantation has been tried.

Further reading

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Articles on endocrinology.

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26

Musculoskeletal disorders

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Features of musculoskeletal disorders in children are:

- Although musculoskeletal presentations are common and often benign and self-limiting, they can be caused by a serious underlying condition such as infection, malignancy or non-accidental injury
- Many concerns of parents about their children's posture are variations of normal alignment in the growing skeleton
- Juvenile idiopathic arthritis (JIA) is the most common cause of chronic arthritis in children.

Assessment of the musculoskeletal system

This should, as a minimum, include the pGALS (paediatric Gait, Arms, Legs, Spine) screen (see p. 25) to identify and localise musculoskeletal problems; any suggestion of a musculoskeletal problem should be followed by more detailed regional musculoskeletal examination (called pREMS, see p. 26).

Variations of normal posture

Variations are common and may be noticed by parents or on routine developmental surveillance. Most resolve without any treatment but if severe, progressive, painful or asymmetrical, they should be referred for specialist opinion.

Bow legs (genu varum)

The normal toddler has a broad base gait. Many children evolve leg alignment with initially a degree of bowing of the tibiae, causing the knees to be wide apart – best observed while the child is standing with the feet together (Fig. 26.1). A pathological cause of bow legs is rickets; check for the presence of other clinical features (see Ch. 11). Severe progressive and often unilateral bow legs is a feature of Blount disease (infantile tibia vara), an uncommon condition predominantly seen in Afro-Caribbean children. Radiographs are characteristic with beaking of the proximal medial tibial epiphysis.

Knock-knees (genu valgum)

The feet are wide apart when standing with the knees held together (Fig. 26.2). It is seen in many young children and usually resolves spontaneously.

Flat feet (pes planus)

Toddlers learning to walk usually have flat feet due to flatness of the medial longitudinal arch and the presence of a fat pad which disappears as the child gets older (Fig. 26.3). An arch can usually be demonstrated on standing on tiptoe or by passively extending the big toe. Marked flat feet is common in hypermobility. A fixed flat foot, often painful, presenting in older children, may indicate a congenital tarsal coalition and requires an orthopaedic opinion. Symptomatic flat feet are often helped with footwear advice and, occasionally, an arch support may be required.

In-toeing and out-toeing

There are three main causes of in-toeing:

- *Metatarsus varus* (Fig. 26.4a) – an adduction deformity of a highly mobile forefoot

Variants of normal

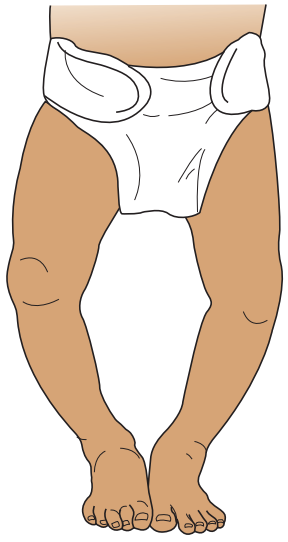


Figure 26.1 Bow legs.

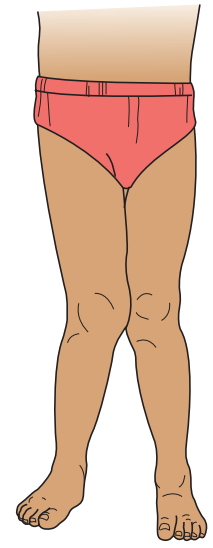


Figure 26.2 Knock-knees.

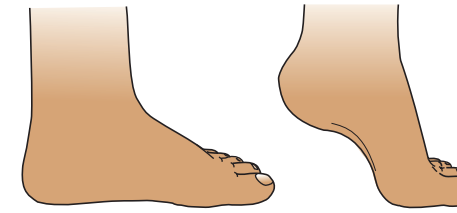


Figure 26.3 Pes planus showing the flat feet of toddlers. The medial longitudinal arch appears on standing on tiptoe.

In-toeing

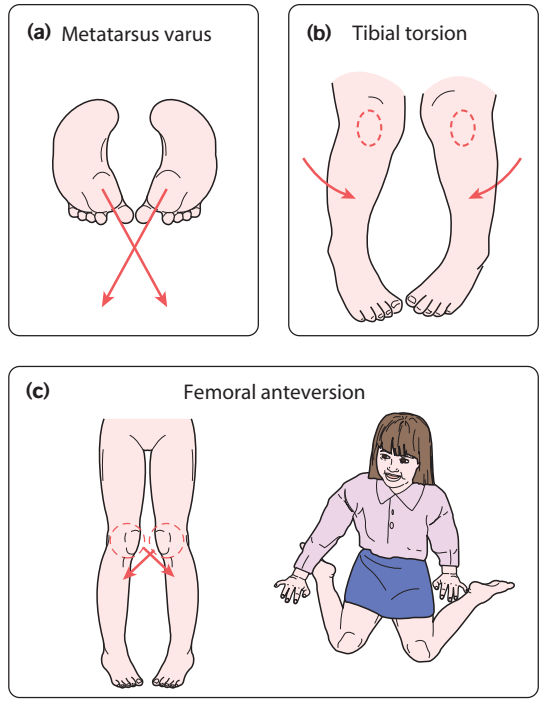


Figure 26.4 In-toeing **(a)** at the feet, **(b)** lower leg, **(c)** hip, with 'W' sitting.

Box 26.1 Clinical features of in-toeing in children

Metatarsus varus

- Occurs in infants
- Passively correctable
- Heel is held in the normal position
- No treatment required unless it persists beyond 5 years of age and is symptomatic

Medial tibial torsion

- Occurs in toddlers
- May be associated with bowing of the tibiae
- Self-corrects within about 5 years

Persistent anteversion of the femoral neck

- Presents in childhood
- Usually self-corrects by 8 years of age
- May be associated with hypermobility of the joints
- Children sit between their feet with the hips fully internally rotated ('W' sitting)
- Most do not require treatment but femoral osteotomy may be required for persistent anteversion.

- *Medial tibial torsion* (Fig. 26.4b) – at the lower leg, when the tibia is laterally rotated less than normal in relation to the femur
- *Persistent anteversion of the femoral neck* (Fig. 26.4c) – at the hip, when the femoral neck is twisted forward more than normal.

The clinical features are described in Box 26.1.

Out-toeing is uncommon but may occur in infants between 6 and 12 months of age.

When bilateral, it is often due to lateral rotation of the hips and resolves spontaneously.

Summary

Variations of musculoskeletal normality and differential diagnosis

Perceived disorder	Normal age range	Differential diagnoses to consider
Bow legs	1–3 years	Rickets osteogenesis imperfecta, Blount disease
Knock-knees	2–7 years	Juvenile idiopathic arthritis (JIA)
Flat feet	1–2 years	Hypermobility, congenital tarsal fusion
In-toeing	1–2 years	Tibial torsion, femoral anteversion
Out-toeing	6–12 months	Hypermobility, Ehlers–Danlos and Marfan syndromes
Toe walking	1–3 years	Spastic diplegia, muscular dystrophy, JIA

Toe walking

Common in young children and may become persistent, usually from habit; can walk normally on request. Needs to be distinguished from mild cerebral palsy or tightness of the Achilles tendons or inflammatory arthritis in the foot or ankle. In older boys, Duchenne muscular dystrophy should be excluded.

Abnormal posture

Talipes equinovarus (clubfoot)

Positional talipes from intrauterine compression is common. The foot is of normal size, the deformity is mild and can be corrected to the neutral position with passive manipulation. Often the baby's intrauterine posture can be recreated. If the positional deformity is marked, parents can be shown passive exercises by the physiotherapist.

Talipes equinovarus is a complex abnormality (Figs 26.5, 26.6). The entire foot is inverted and supinated, the forefoot adducted and the heel is rotated inwards and in plantar flexion. The affected foot is shorter and the calf muscles thinner than normal. The position of the foot is fixed, cannot be corrected completely and is often bilateral. The birth prevalence is 1 per 1000 live births, affects predominantly males (2:1), can be familial and is usually idiopathic. However, it may also be secondary to oligohydramnios during pregnancy, a feature of a malformation syndrome or of a neuromuscular disorder such as spina bifida. There is an association with developmental dysplasia of the hip (DDH).

Treatment is started promptly with plaster casting and bracing ('Ponsetti method'), which may be required for many months. It is usually successful unless the condition is very severe, when corrective surgery is required.

Vertical talus

Talipes equinovarus needs to be differentiated from the rare congenital vertical talus, where the foot is stiff and rocker-bottom in shape. Many of these infants have other malformations. The diagnosis can be confirmed on X-ray. Surgery is usually required.

Talipes

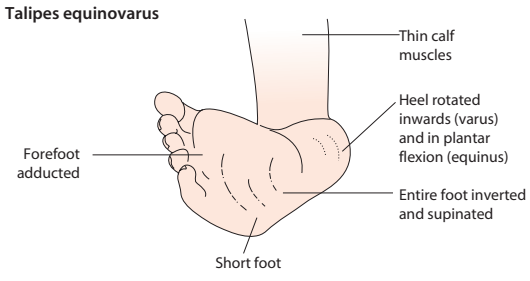


Figure 26.5 Abnormalities in talipes equinovarus.



Figure 26.6 Talipes equinovarus.

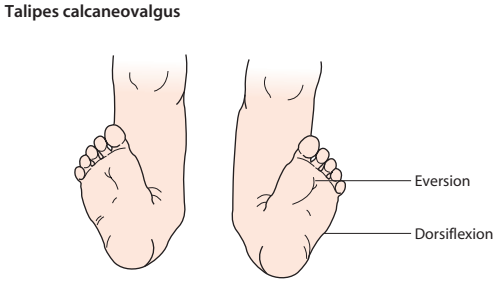


Figure 26.7 Talipes calcaneovalgus.

Talipes calcaneovalgus

The foot is dorsiflexed and everted (Fig. 26.7). It usually results from intrauterine moulding and self-corrects. Passive foot exercises are sometimes advised.

There is an association with developmental dysplasia of the hip.

Flat feet

In older children and adolescents, a *rigid* flat foot is pathological. It is suggested by absence of a normal arch on tip toeing. It may be due to an associated tendo-Achilles contracture (ankle), or tarsal coalition (see below) or inflammatory arthropathy (juvenile idiopathic arthritis, JIA).

Tarsal coalition

Results from lack of segmentation between one or more bones of the foot, and coalitions that were fibrous or cartilaginous become symptomatic as they begin to ossify. They become progressively more rigid and limit normal foot motion. They often become symptomatic during the pre-adolescent years. Radiographs may be normal if the bars have not yet ossified. Corrective surgery may be required.

Pes cavus

In pes cavus, there is a high arched foot. When it presents in older children, it is often associated with neuromuscular disorders, e.g. Friedreich ataxia and type I hereditary motor sensory neuropathy (peroneal muscular atrophy). Treatment is required if the foot becomes stiff or painful.

Summary

Regarding talipes equinovarus

- Needs to be differentiated from positional talipes
- Check for neuromuscular disorder or spinal lesion and for developmental dysplasia of the hip (DDH).

Developmental dysplasia of the hip (DDH)

This is a spectrum of disorders ranging from dysplasia to subluxation through to frank dislocation of the hip. Early detection is important as it usually responds to conservative treatment; late diagnosis is usually associated with hip dysplasia, which requires complex treatment often including surgery. Neonatal screening is performed as part of the routine examination of the newborn (see Fig. 9.15), checking if the hip can be dislocated posteriorly out of the acetabulum (Barlow manoeuvre) or can be relocated back into the acetabulum on abduction (Ortolani manoeuvre). These tests are repeated at routine surveillance at 8 weeks of age. Thereafter, presentation of the condition is usually with a limp or abnormal gait. It may be identified from asymmetry of skinfolds around the hip, limited abduction of the hip or shortening of the affected leg.

On neonatal screening, an abnormality of the hip is detected in about 6–10 per 1000 live births. Most will resolve spontaneously. The true birth prevalence of DDH is about 1.3 per 1000 live births. Clinical neonatal screening misses some cases. This may be because of inexperience of the examiner, but in some it is not possible to clinically detect dislocation at this stage, e.g. where there is only a mildly shallow acetabulum. To overcome these problems, some centres perform ultrasound screening on all newborn infants. It is highly specific in detecting DDH but is expensive and has a high rate of false positives, and is not recommended in the UK. It is performed in some centres in infants at increased risk (family history, of breech presentation).

If developmental dysplasia of the hip is suspected, a specialist orthopaedic opinion should be obtained. An ultrasound examination allows detailed assessment of the hip, quantifying the degree of dysplasia and whether there is subluxation or dislocation. If the initial ultrasound is abnormal, the infant may be placed in a splint or harness to keep the hip flexed and abducted for several months. Progress is monitored by ultrasound or X-ray. The splinting must be done expertly as necrosis of the femoral head is a potential complication.

In most instances, a satisfactory response is obtained. Surgery is required if conservative measures fail.

Scoliosis

Scoliosis is a lateral curvature in the frontal plane of the spine.

In structural scoliosis, there is rotation of the vertebral bodies which causes a prominence in the back from rib asymmetry. In most cases, the changes are mild, pain-free and primarily a cosmetic problem; however, in severe cases, the spinal curvature can lead to cardiorespiratory failure from distortion of the chest.

Causes of scoliosis are:

- *Idiopathic*. The most common, either early onset (<5 years old) or, most often, late onset, mainly girls 10–14 years of age during their pubertal growth spurt
- *Congenital*. From a congenital structural defect of the spine, e.g. hemivertebra, spina bifida, syndromes (e.g. VACTERL association – Vertebral, Anorectal, Cardiac, Tracheo-oEsophageal, Renal and Limb anomalies).
- *Secondary*. Related to other disorders such as neuromuscular imbalance (e.g. cerebral palsy, muscular dystrophy); disorders of bone such as neurofibromatosis or of connective tissues such as Marfan syndrome, or leg length discrepancy, e.g. due to arthritis of one knee in JIA.

Examination should start with inspection of the child's back while standing up straight. In mild scoliosis, there may be irregular skin creases and difference in shoulder

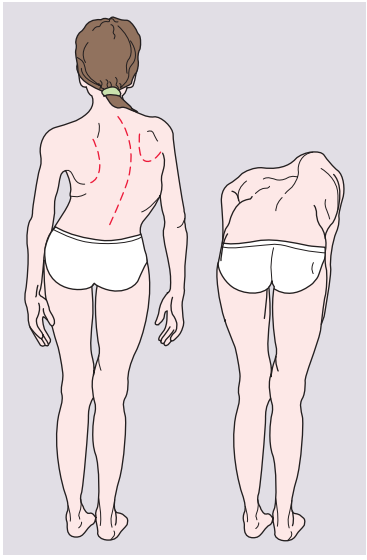


Figure 26.8 Structural scoliosis with vertebral rotation shown by rib rotation on bending forward.

height. The scoliosis can be identified on examining the child's back when bent forward (Fig. 26.8). If the scoliosis disappears on forward bending, it is postural although leg lengths should be checked.

Mild scoliosis will resolve spontaneously, or progresses minimally. If more severe, the severity and progression of the curvature of the spine is determined by X-ray. Severe cases are managed in specialist spinal centres where the place of non-medical treatment such as bracing will be considered, with surgery indicated only if severe or there is coexisting pathology such as neuromuscular or respiratory disease.

Torticollis

The most common cause of torticollis (wry neck) in infants is a sternomastoid tumour (congenital muscular torticollis). They occur in the first few weeks of life and present with a mobile, non-tender nodule, which can be felt within the body of the sternocleidomastoid muscle. There may be restriction of head turning and tilting of the head. The condition usually resolves in 2–6 months. Passive stretching is advised, but its efficacy is unproven.

Torticollis presenting later in childhood may be due to muscular spasm or secondary to ENT infection, spinal tumour (such as osteoid osteoma), cervical spine arthritis or malformation or posterior fossa tumour.

The painful limb, knee and back

Growing pains

Episodes of generalised pain in the lower limbs, referred to as 'growing pains' or nocturnal idiopathic pain, are common in preschool and school-aged children. The pain often wakes the child from sleep and settles with massage or comforting. The condition is poorly understood, but features to be fulfilled for this diagnosis are:

- Age range 3–12 years
- Pains symmetrical in lower limbs and not limited to joints
- Pains *never* present at the start of the day after waking
- Physical activities not limited; no limp
- Physical examination normal and otherwise well (with the exception of joint hypermobility in some).



Figure 26.9 Hypermobility syndrome, showing ability to hyperextend the thumb onto the forearm of a mother and two of her children.

Hypermobility

Older children or adolescents with hypermobility may complain of musculoskeletal pain mainly confined to the lower limbs, often worse after exercise. Joint swelling is usually absent or is transient. Hypermobility may be generalised or limited to peripheral joints (such as hands and feet). There is symmetrical hyperextension of the thumbs and fingers that can be hyperextended onto the forearms (Fig. 26.9), elbows and knees can be hyperextended beyond 10°, and palms can be placed flat on the floor with knees straight. Lower limb findings associated with hypermobility are hyperextensibility of the knee joint and flat feet with normal arches on tiptoe, which are over-pronated secondary to ankle hypermobility.

While mild degrees of hypermobility are a normal finding in younger female children, and many children with hypermobility are asymptomatic and find being very flexible an advantage in dancing and gymnastics, some experience recurrent mechanical joint and muscle pain, which is often activity related. These children require specialist assessment and may benefit from advice about footwear, exercises and occasionally orthotics. Hypermobility is also a feature of some chromosomal syndromes, e.g. Down syndrome and some inherited collagen disorders (e.g. Marfan and Ehlers–Danlos syndrome).

Complex regional pain syndromes

The most dramatic musculoskeletal pain is that encountered in complex regional pain syndromes (CRPS), formerly known as idiopathic pain syndromes, which may

be localised or generalised. They usually present in adolescent females.

Localised forms often present with foot and ankle involvement (typically unilateral); the pain can be extreme and incapacitating, often triggered by minor trauma or without a clear precipitant. Presentation to the clinic may be in a wheelchair. In addition to severe pain, there may be hyperaesthesia (increased sensitivity to stimuli), allodynia (pain from a stimulus that does not normally produce pain), and the affected part (often a foot or hand) may be cool to touch with swelling and mottling, held in flexion with minimal if any active movement, and bizarre posturing is not uncommon. Typically, with distraction, the normal range of passive movements is possible.

Diffuse forms are characterised by severe widespread pain with disturbed sleep patterns, feeling exhausted during the day, with extreme tenderness over soft tissues. The characteristic tender points that are found in adults with fibromyalgia may be absent or fewer in number in children.

The child or adolescent with complex regional pain is otherwise well and physical examination is otherwise normal.

Organic pathology needs to be excluded. The aetiology is unknown, but affected children often have significant associated stresses in their lives.

A multidisciplinary rehabilitation regimen is required, predominantly physical therapy-based, either community or inpatient.

Acute-onset limb pain

Limb pain of acute onset has a number of causes. Trauma is the most common, usually accidental from sports injuries or falls, but occasionally non-accidental. Osteomyelitis and bone tumours are uncommon, but need urgent treatment.

Osteomyelitis

In osteomyelitis, there is infection of the metaphysis of long bones. The most common sites are the distal femur and proximal tibia, but any bone may be affected (Fig. 26.10). It is usually due to haematogenous spread of the pathogen, but may arise by direct spread from an infected wound. The skin is swollen directly over the affected site. Where the joint capsule is inserted distal to the epiphyseal plate, as in the hip, osteomyelitis may spread to cause septic arthritis. Most infections are caused by *Staphylococcus aureus*, but other pathogens include *Streptococcus* and *Haemophilus influenzae* if not immunised. In sickle cell anaemia, there is an increased risk of staphylococcal and salmonella osteomyelitis. Infection may be from tuberculosis; although rare in the UK, it needs to be considered, especially in the immunodeficient child.

Presentation

This is usually with a markedly painful, immobile limb (pseudoparesis) in a child with an acute febrile illness. Directly over the infected site there is swelling and exquisite tenderness, and it may be erythematous and warm. Moving the limb causes severe pain. There may be a sterile effusion of an adjacent joint. Presentation may be more insidious in infants, in whom swelling or reduced limb movement is the initial sign. Beyond infancy, presentation may be with back pain in a vertebral infection or with a limp or groin pain in infection of the pelvis. Occasionally, there are multiple foci (e.g. disseminated staphylococcal or *H. influenzae* infection).

Investigation

Blood cultures are usually positive and the white blood count and acute-phase reactants are raised. X-rays are initially normal, other than showing soft tissue swelling; it takes 7–10 days for subperiosteal new bone formation and localised bone rarefaction to become visible. Ultrasound may show periosteal elevation at presentation. MRI allows identification of infection in the bone (subperiosteal pus and purulent debris in the bone) and differentiation of bone from soft tissue infection. Radionuclide bone scan (Fig. 26.11) may be helpful if the site of infection is unclear. The X-ray changes of chronic osteomyelitis are shown in Figure 26.12.

Treatment

Prompt treatment with parenteral antibiotics is required for several weeks to prevent bone necrosis, chronic infection with a discharging sinus, limb deformity and amyloidosis. Antibiotics are given intravenously until there is clinical recovery and the acute-phase reactants have returned to normal, followed by oral therapy for several weeks. Aspiration or surgical decompression of the subperiosteal space may be performed if the presentation is atypical or in immunodeficient children. Surgical drainage is performed if the condition does not respond rapidly to antibiotic therapy. The affected limb is initially rested in a splint and subsequently mobilised.

Summary

Osteomyelitis

- Presents with fever, a painful, immobile limb, swelling and extreme tenderness, especially on moving the limb
- Blood cultures are usually positive
- Parenteral antibiotics must be given immediately
- Surgical drainage if unresponsive to antibiotic therapy.

Malignant disease

Acute lymphoblastic leukaemia may present with bone pain in children (sometimes primarily at night) and even frank arthritis. Neuroblastoma, usually in the young children, may present with systemic arthritis, or bone pain from metastases, which may be difficult to localise.

Bone tumours

Malignant bone tumours – osteogenic sarcoma and Ewing tumour – are rare. They present with pain or

Osteomyelitis

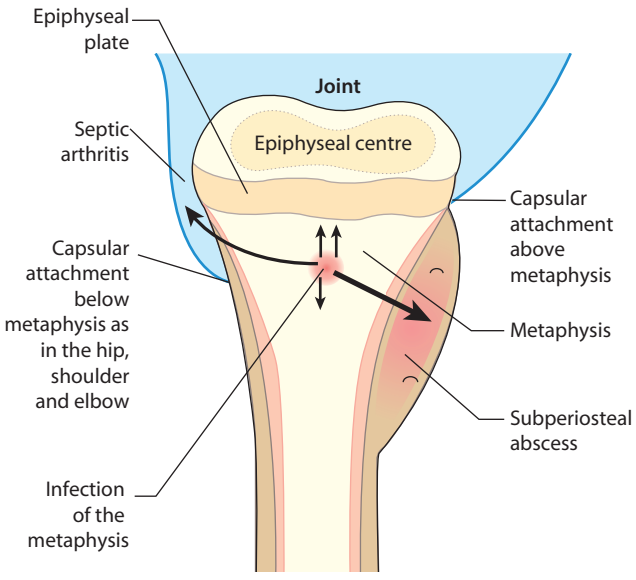


Figure 26.10 Possible spread of osteomyelitis. In children, the epiphyseal growth plate limits the spread of metaphyseal infection. In infants, before there has been maturation of the growth plate, infection can spread directly to cause joint destruction and arrested growth.

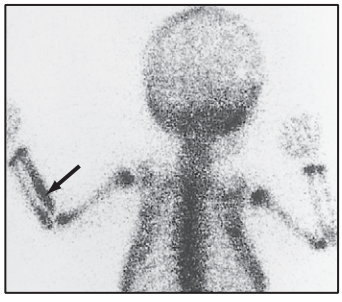


Figure 26.11 Bone scan of osteomyelitis with increased radionuclide uptake of the left radius. (Courtesy of Professor H. Carty.)



Figure 26.12 Chronic osteomyelitis, showing periosteal reaction along the lateral shaft of the tibia and multiple hypodense areas within the metaphyseal regions.

swelling, or occasionally with a pathological fracture. Further features are considered in Chapter 21.

Osteoid osteoma is a benign tumour affecting adolescents, especially boys, usually involving the femur or tibia or spine. The pain is more severe at night and improves with NSAID therapy. There may be some localised tenderness, soft tissue swelling, joint effusion if sited near a joint and scoliosis if in the spine. The X-ray is usually diagnostic, with a sharply demarcated radiolucent nidus of osteoid tissue surrounded by sclerotic bone. If the X-ray is normal, a CT or MRI scan is required. Treatment is by surgical removal.

The painful knee

When assessing a painful knee, the hip must always be examined, as hip pain is often referred to the knee.

Osgood–Schlatter disease

This is osteochondritis of the patellar tendon insertion at the knee, often affecting adolescent males who are physically active (particularly football or basketball). Usually presents with knee pain after exercise, localised tenderness and sometimes swelling over the tibial tuberosity. There is often hamstring tightness. It is bilateral in 25–50%. Most resolve with reduced activity and physiotherapy for quadriceps muscle strengthening, hamstring stretches and occasionally orthotics. A knee immobiliser splint may be helpful.

Chondromalacia patellae

There is softening of the articular cartilage of the patella. It most often affects adolescent females, causing pain when the patella is tightly apposed to the femoral condyles, as in standing up from sitting or on

walking up stairs. It is often associated with hypermobility and flat feet, suggesting a biomechanical component to the aetiology

Treatment is with rest and physiotherapy for quadriceps muscle strengthening.

Osteochondritis dissecans (segmental avascular necrosis of the subchondral bone)

This presents as persistent knee pain in the physically very active adolescent, with localised tenderness over the femoral condyles. Pain is caused by separation of bone and cartilage from the medial femoral condyle following avascular necrosis. Complete separation of articular fragments may result in loose body formation and symptoms of knee locking or giving way. Treatment is initially with rest and quadriceps exercises; sometimes arthroscopic surgery is required.

Subluxation and dislocation of the patella

Subluxation of the patella produces the feeling of instability or giving way of the knee. It is often associated with generalised hypermobility. Rarely, dislocation of the patella can occur, usually laterally, suddenly and with severe pain – reduction occurs spontaneously or on gentle extension of the knee. Treatment is with quadriceps exercises. Sometimes surgery is required to realign the pull of the quadriceps on the patellar tendon.

Injuries

Contact sports characteristically result in acute injuries to the knee, while non-contact sports with sustained activity tend to result in chronic injury and overuse syndromes. Sporting injuries to the menisci and ligaments are common in adolescents. MRI scans are helpful to determine the extent of damage. Management is usually conservative. In infants and young children, similar injuries are more likely to result in fractures, as their ligaments are relatively stronger than their bones.

Back pain

Back pain is a symptom of concern in the very young and pre-adolescent ages as, in contrast to adults, a cause can often be identified. The younger the child, the more likely there will be significant pathology. Red Flag clinical features are listed in Box 26.2.

Mechanical causes – there may be muscle spasm *or* soft tissue pain from injury, often sport-related or from poor posture or abnormal loading (such as carrying heavy school bags on one shoulder).

Tumours: benign or malignant – The spine is a common site for osteoid osteoma. It may also be the site of primary tumours or metastases.

Vertebral osteomyelitis or discitis – there is localised tenderness; in infants there is reluctance to walk or bear weight or pain on spine flexion along with fever and systemic upset. While plain X-rays may show abnormalities suggesting the diagnosis, further imaging (MRI) is often required. Treatment is with intravenous antibiotics.

Box 26.2 Red Flag clinical features of back pain

- Young age – pathology more likely
- High fever – infection
- Night waking, persistent pain – osteoid osteoma or tumours
- Painful scoliosis – infection or malignancy
- Focal neurological signs including nerve root irritation, loss of bowel/bladder control – nerve root/spinal cord compression
- Associated weight loss, systemic malaise – malignancy.

Spinal cord or nerve root entrapment – from tumour or prolapsed intervertebral disc – often associated with trauma or heavy lifting,

Scheuermann disease – an osteochondrosis of the vertebral body; may present with a fixed thoracic kyphosis with or without back pain. The diagnosis is usually made on X-ray. In many cases, the radiographic changes are a coincidental finding and the patient is asymptomatic.

Spondylolysis/spondylolisthesis – stress fracture of the pars interarticularis of the vertebra. Increased risk with certain sporting activities, e.g. bowling in cricket or gymnastics. If bilateral, can result in spondylolisthesis, forward slip of the vertebral body and potential cord or nerve root compression. There is pain on spine extension and localised tenderness. Change may be apparent on X-ray but often further imaging (CT scan) is required.

Complex regional pain syndrome (CRPS) – diagnosed when no physical cause is found; may be exacerbated by psychological stress.

Limp

Limp can be divided into acute painful limp and chronic or intermittent limp, where pain may or may not be the presenting feature, and by age (Table 26.1).

Transient synovitis ('irritable hip')

This is the most common cause of acute hip pain in children. It occurs in children aged 2–12 years old. It often follows or is accompanied by a viral infection. Presentation is with sudden onset of pain in the hip or a limp. There is no pain at rest, but there is decreased range of movement, particularly internal rotation. The pain may be referred to the knee. The child is afebrile or has a mild fever and does not appear ill.

It can be difficult to differentiate transient synovitis from early septic arthritis of the hip joint (Table 26.2), and if there is any suspicion of septic arthritis, joint aspiration and blood cultures are mandatory.

In a small proportion of children, transient synovitis precedes the development of Perthes disease. Management of transient synovitis is with bed rest

Table 26.1 Causes of limp

	Acute painful limp	Chronic and intermittent limp
1–3 years	Infection – septic arthritis, osteomyelitis of hip or spine Transient synovitis Trauma – accidental/non-accidental Malignant disease – leukaemia, neuroblastoma	Developmental dysplasia of the hip (DDH), talipes Neuromuscular, e.g. cerebral palsy Juvenile idiopathic arthritis (JIA)
3–10 years	Transient synovitis Septic arthritis/osteomyelitis Trauma and overuse injuries Perthes disease (acute) Juvenile idiopathic arthritis (JIA) Malignant disease, e.g. leukaemia Complex regional pain syndrome	Perthes disease (chronic) Neuromuscular disorders, e.g. Duchenne muscular dystrophy Juvenile idiopathic arthritis (JIA) Tarsal coalition
11–16 years	Mechanical – trauma, overuse injuries, sport injuries Slipped capital femoral epiphysis (acute) Avascular necrosis of the femoral head Reactive arthritis Juvenile idiopathic arthritis (JIA) Septic arthritis/osteomyelitis Osteochondritis dissecans of the knee Bone tumours and malignancy Complex regional pain syndrome	Slipped capital femoral epiphysis (chronic) Juvenile idiopathic arthritis (JIA) Tarsal coalition

Table 26.2 Contrast in clinical features of transient synovitis and septic arthritis of the hip

	Transient synovitis	Septic arthritis
Onset	Acute limp, non-weight bearing	Acute onset, non-weight bearing
Fever	Mild/absent	Moderate/high
Child's appearance	Child often looks well	Child looks ill
Hip movement	Comfortable at rest, limited internal rotation and pain on movement	Hip held flexed; severe pain at rest and worse on any attempt to move joint
White cell count	Normal	Normal/high
Acute-phase reactant/ESR	Slight increase/normal	Raised
Ultrasound	Fluid in joint	Fluid in joint
Radiograph	Normal	Normal/widened joint space
Management	Rest, analgesia	Joint aspiration, usually under ultrasound guidance Prolonged antibiotics, rest and analgesia
Course	Resolves <1 week, approx 3% develop Perthes disease	Progressive and severe joint damage if not treated

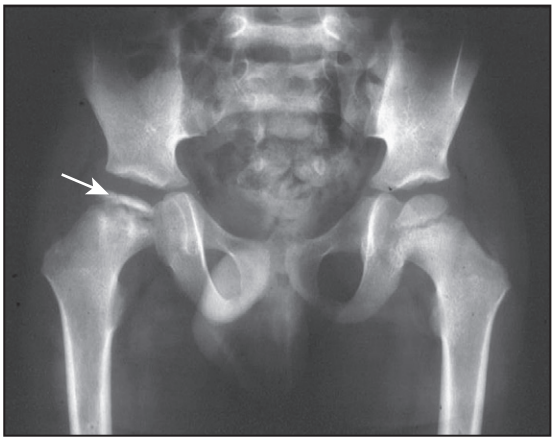


Figure 26.13 Perthes disease, showing flattening with sclerosis and fragmentation of the right femoral capital epiphysis; the left hip is normal.

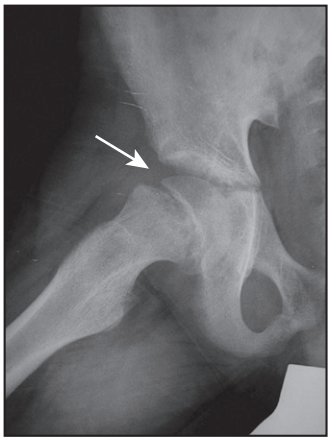


Figure 26.14 Slipped capital femoral epiphysis of the right hip.

and, rarely, skin traction. It usually improves within a few days.

Perthes disease

This is an avascular necrosis of the capital femoral epiphysis of the femoral head due to interruption of the blood supply, followed by revascularisation and reossification over 18–36 months. It mainly affects boys (male : female ratio of 5 : 1) of 5–10 years of age. Presentation is insidious, with the onset of a limp, or hip or knee pain. The condition may initially be mistaken for transient synovitis. It is bilateral in 10–20%. If suspected, X-ray of both hips (including frog views) should be requested; early signs of Perthes include increased density in the femoral head, which subsequently becomes fragmented and irregular (Fig. 26.13).

Even if the initial X-ray is normal, a repeat may be required if clinical symptoms persist. A bone scan and MRI scan can be helpful in making the diagnosis.

Prognosis is dependent on early diagnosis; if identified early and less than half the femoral head is affected, only bed rest and traction may be required. In more severe disease or late presentations, the femoral head needs to be covered by the acetabulum to act as a mould for the re-ossifying epiphysis and is achieved by maintaining the hip in abduction with plaster or calipers, or by performing femoral or pelvic osteotomy.

In most children, the prognosis is good, particularly in those below 6 years of age with less than half the epiphysis involved. In older children or with more extensive involvement of the epiphysis, deformity of the femoral head and metaphyseal damage are more likely, with potential for subsequent degenerative arthritis in adult life.

Slipped capital femoral epiphysis (SCFE)

Results in displacement of the epiphysis of the femoral head postero-inferiorly requiring prompt treatment in order to prevent avascular necrosis. It is most common at 10–15 years of age during the adolescent growth spurt, particularly in obese boys and is bilateral in 20%. There is an association with metabolic endocrine abnormalities, e.g. hypothyroidism and hypogonadism. Presentation is with a limp or hip pain, which may be referred to the knee. The onset may be acute, following minor trauma or insidious. Examination shows restricted abduction and internal rotation of the hip. Diagnosis is confirmed on X-ray (Fig. 26.14), and a frog lateral view should also be requested. Management is surgical, usually with pin fixation in situ.

Summary

Regarding hip disorders

- Developmental dysplasia of the hip – identified on screening at birth or 8 weeks, detection of asymmetry of skinfolds around the hip, limited abduction of the hip, shortening of the affected leg or a limp or abnormal gait
- Transient synovitis – most common cause of acute hip pain or a limp; must be differentiated from septic arthritis
- Perthes disease – usually school-aged children with hip pain or limp
- Slipped capital femoral epiphysis – adolescent with a limp or hip pain.

Arthritis

Acute arthritis presents with pain, swelling, heat, redness and restricted movement in a joint. In a monoarthritis of acute onset, the child is also likely to be systemically unwell with fever; if septic arthritis or osteomyelitis is the cause, urgent diagnosis and treatment is required. With infection, more than one joint can be affected, although a single joint is more common.

The causes of polyarthritis are listed in Table 26.3.

Table 26.3 Causes of polyarthritis

Infection	Bacterial – septicaemia/septic arthritis, TB
	Viral – rubella, mumps, adenovirus, coxsackie B, herpes, hepatitis, parvovirus
	Other – *Mycoplasma*, Lyme disease, rickettsia
	Reactive – gastrointestinal infection, streptococcal infection
	Rheumatic fever
Inflammatory bowel disease	Crohn disease, ulcerative colitis
Vasculitis	Henoch–Schönlein purpura, Kawasaki disease
Haematological disorders	Haemophilia, sickle cell disease
Malignant disorders	Leukaemia, neuroblastoma
Connective tissue disorders	Juvenile idiopathic arthritis (JIA), systemic lupus erythematosus (SLE), dermatomyositis, mixed connective tissue disease (MCTD), polyarteritis nodosa (PAN)
Other	Cystic fibrosis

Reactive arthritis

Reactive arthritis is the most common form of arthritis in childhood. It is characterised by transient joint swelling (usually <6 weeks) often of the ankles or knees. It usually follows (or rarely accompanies) evidence of extra-articular infection. The enteric bacteria (*Salmonella*, *Shigella*, *Campylobacter* and *Yersinia*) are often the cause in children, but viral infections, sexually transmitted infections in adolescents (chlamydia, gonococcus), *Mycoplasma* and *Borrelia burgdorferi* (Lyme disease) are other causes. Rheumatic fever and post-streptococcal reactive arthritis are rare in developed countries but are frequent in many developing countries.

Fever is low grade. Acute-phase reactants are normal or mildly elevated and X-rays are normal. No treatment or only NSAIDs are required and complete recovery can be anticipated.

Septic arthritis

This is a serious infection of the joint space, as it can lead to bone destruction. It is most common in children <2 years old. It usually results from haematogenous spread, but may also occur following a puncture wound or infected skin lesions, e.g. chickenpox. In young children, it may result from spread from adjacent osteomyelitis into joints where the capsule inserts below the epiphyseal growth plate. Usually only one joint is affected, with the hip being a particular concern in infants and young children. Beyond the neonatal period, the most common organism is *Staphylococcus aureus*, and usually only one joint is affected. *H. influenzae* was an important cause in young children prior to Hib immunisation and often affected multiple sites. Underlying and predisposing illnesses such as immunodeficiency and sickle cell disease should be considered.

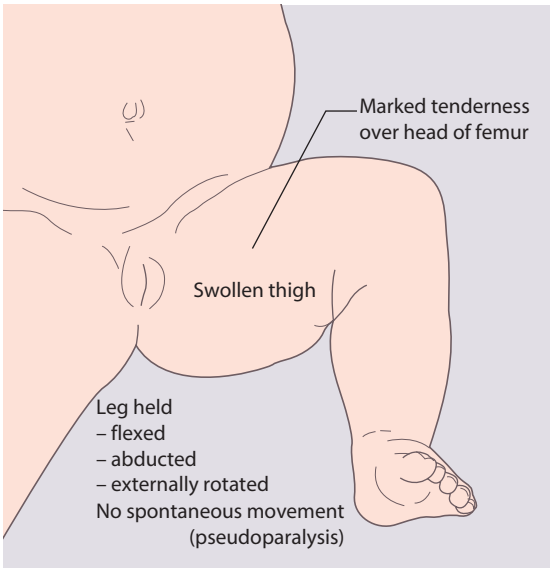


Figure 26.15 Septic arthritis of the hip in infants, showing the characteristic posture to reduce intracapsular pressure. Any leg movement is painful and is resisted.

Presentation

This is usually with an erythematous, warm, acutely tender joint, with a reduced range of movement, in an acutely unwell, febrile child. Infants often hold the limb still (pseudoparesis, pseudoparalysis) and cry if it is moved. A joint effusion may be detectable in peripheral joints. In osteomyelitis, although a sympathetic joint effusion may be present, the tenderness is over the bone, but in up to 15% there is coexistent septic arthritis. The diagnosis of septic arthritis of the hip can be particularly difficult in toddlers, as the joint is well covered by subcutaneous fat (Fig. 26.15). Initial presentation may be with a limp or pain referred to the knee.

Investigation

There is an increased white cell count and acute-phase reactants. Blood cultures must be taken. Ultrasound of deep joints, such as the hip, is helpful to identify an effusion. X-rays are used to exclude trauma and other bony lesions. However, in septic arthritis, the X-rays are initially normal, apart from widening of the joint space

and soft tissue swelling. A bone scan may be helpful and an MRI scan may demonstrate an adjacent osteomyelitis. Aspiration of the joint space under ultrasound guidance for organisms and culture is the definitive investigation. Ideally, this is performed immediately, unless this would cause a significant delay in giving antibiotics. A prolonged course of antibiotics is required, initially intravenously. Washing out of the joint or surgical drainage may be required if resolution does not occur rapidly or if the joint is deep-seated, such as the hip. The joint is initially immobilised in a functional position, but subsequently must be mobilised to prevent permanent deformity.

Early treatment of septic arthritis is essential to prevent destruction of the articular cartilage and bone.

Juvenile idiopathic arthritis (JIA)

This is the commonest chronic inflammatory joint disease in children and adolescents in the UK. It is defined as persistent joint swelling (of >6 weeks duration) presenting before 16 years of age in the absence of infection or any other defined cause. Ninety-five per cent of children have a disease that is clinically and immunogenetically distinct from rheumatoid arthritis in adults. It has a prevalence of approximately 1 in 1000 children, (i.e. similar to epilepsy), with over 12 000 affected children in the UK.

There are at least seven different subtypes of JIA. Its classification is clinical and based on the number of joints affected in the first 6 months, as polyarthritis (more than four joints) (Fig. 26.16) and oligoarthritis (up to and including four joints) or systemic (with fever and rash). Psoriatic arthritis and enthesitis are further subtypes. Subtyping is further classified according to the presence of rheumatoid factor and HLA B27 tissue type. The subtypes and their clinical features are shown in Table 26.4.

Features in the history are gelling (stiffness after periods of rest, such as long car rides), morning joint stiffness and pain. In the young child, it may present with intermittent limp or deterioration in behaviour or mood or avoidance of previously enjoyed activities, rather than complaining of pain.

Initially, there may be only minimal evidence of joint swelling, but subsequently there may be swelling of the joint due to fluid within it, inflammation and, in chronic arthritis, proliferation (thickening) of the synovium and swelling of the periarticular soft tissues.

Long term, with uncontrolled disease activity, there may be bone expansion from overgrowth, which in the knee may cause leg lengthening or valgus deformity, in the hands, discrepancy in digit length, and in the wrist, advancement of bone age.

If systemic features are present, sepsis and malignancy must always be considered.

Complications

Chronic anterior uveitis

This is common but asymptomatic and can lead to severe visual impairment. Regular ophthalmological screening using a slit lamp is indicated, especially for children with oligoarticular disease.



Figure 26.16 Polyarticular juvenile idiopathic arthritis, showing swelling of the wrists, metacarpal and interphalangeal joints and early swan-neck deformities of the fingers.



Figure 26.17 Growth failure and marked genu valgum (knock-knees) in an 8-year-old girl with juvenile idiopathic arthritis. For comparison, her sister on the left is 4 years old.

Flexion contractures of the joints

These occur when the joint is held in the most comfortable position, thereby minimising intra-articular pressure. Chronic untreated disease can lead to joint destruction and the need for joint replacement.

Growth failure

This may be generalised from anorexia, chronic disease and systemic corticosteroid therapy (Fig. 26.17). May also be localised overgrowth such as leg length discrepancy due to prolonged active knee synovitis and undergrowth, such as micrognathia, usually seen in

Table 26.4 Classification and clinical features of JIA (juvenile idiopathic arthritis)

JIA subtype (approximate %)	Onset age	Sex ratio (F:M)	Articular pattern	Extra-articular features	Laboratory abnormalities
Oligoarthritis (persistent) (49%)	1–6 years	5:1	1–4 (max) joints involved; knee, ankle or wrist most common	Chronic anterior uveitis in 20%, leg length discrepancy Prognosis excellent	ANA+/–
Oligoarthritis (extended) (8%)	1–6 years	5:1	>4 joints involved after first 6 months. Asymmetrical distribution of large and small joints	Chronic anterior uveitis 20%, asymmetrical growth Prognosis moderate	ANA+/–
Polyarthritis (RF negative) (16%)	1–6 years	5:1	Symmetrical large and small joint arthritis, often with marked finger involvement Cervical spine and temporomandibular joint may be involved	Low-grade fever, chronic anterior uveitis 5%, late reduction of growth rate Prognosis moderate	
Polyarthritis (RF) positive) (3%)	10–16 years	5:1	Symmetrical large and small joint arthritis, often with marked finger involvement	Rheumatoid nodules 10% Similar to adult rheumatoid arthritis Prognosis poor	RF+ (long term)
Systemic arthritis (9%)	1–10 years	1:1	Oligoarthritis or polyarthritis. May have aches and pains in joints and muscles (arthralgia/myalgia) but initially no arthritis	Acute illness, malaise, high daily fever initially, with salmon-pink, macular rash, lymphadenopathy, hepatosplenomegaly, serositis Prognosis variable to poor	Anaemia, raised neutrophils and platelets, high acute-phase reactants (see Case History 26.1)
Psoriatic arthritis (7%)	1–16 years	1:1	Usually asymmetrical distribution of large and small joints, dactylitis	Psoriasis, nail pitting or dystrophy, chronic anterior uveitis 20% Prognosis moderate	
Enthesitis-related arthritis (7%)	6–16 years	1:4	Lower limb, large joint arthritis initially, mild lumbar spine or sacroiliac involvement later on	Enthesitis – localised inflammation at insertion of tendons or ligaments into bone, often in feet, Achilles insertion Occasional acute uveitis Prognosis moderate	HLAB27+
Undifferentiated arthritis (1%)	1–16 years	2:1 (variable)	Overlapping articular and extra-articular patterns between ≥2 subtypes or insufficient criteria for sub-classification	Prognosis variable	

Case History

26.1 Systemic-onset juvenile idiopathic arthritis

A 2-year-old boy presented with a high fever (Fig. 26.18a) and malaise. A salmon-coloured rash was present at times of fever (Fig. 26.18b). Investigation showed markedly raised acute-phase reactants. Shortly afterwards, he developed severe polyarthritic joint disease. A diagnosis of systemic-onset juvenile idiopathic arthritis was made on the basis of the clinical presentation and exclusion of other disorders (Table 26.3).

He was treated with high-dose intravenous corticosteroids with rapid improvement, started on oral corticosteroids and weekly methotrexate given by subcutaneous injection. His mother was taught by the nurse specialist how to give the injections to him at home and a daily exercise programme to improve his mobility was provided. A year later he remained on weekly methotrexate. He had persistent problems with his hips, with joint damage on X-ray and may ultimately require hip replacements in his adult years. He is shorter than his peers. He is now at university studying economics and drives his own car.

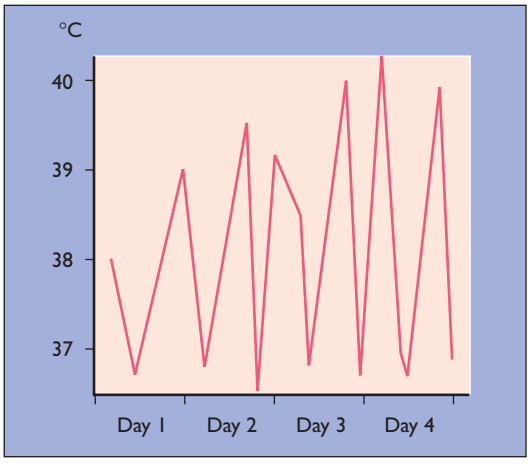


Figure 26.18a Temperature chart.



Figure 26.18b Salmon-pink rash.

long-standing or suboptimally treated arthritis due to premature fusion of epiphyses.

Constitutional problems

Anaemia of chronic disease, delayed puberty.

Osteoporosis

Multifactorial aetiology, including diet, reduced weight bearing, systemic corticosteroids and delayed menarche. Reduce risk by dietary supplements of calcium and vitamin D; regular weight-bearing exercise; and minimise oral corticosteroids use and sometimes bisphosphonates.

Amyloidosis

Very rare now, causes proteinuria and subsequent renal failure and has a high mortality.

Management

The management of JIA has radically changed in the last decade and improvement in outcome is evident as long as children access appropriate care. Deformity and disability are much less common with current treatment approaches. The overall management aim is to induce remission as soon as possible.

All children suspected of having JIA should be managed by specialist paediatric rheumatology multidisciplinary teams, often working in shared care with local hospitals; such teams have specific paediatric expertise in the use and monitoring of immunosuppressive treatments that are now routinely used. There is need for education and support for the child and family, physical therapy to maintain joint function, and links to other specialities including ophthalmology, dentistry and orthopaedics. The team work closely with school, social services and primary healthcare providers. The child is encouraged to take part in all activities except contact sports during active flares. With optimal care, most children are managed as outpatients.

Medical management includes:

- *NSAIDs (non-steroidal anti-inflammatory drugs) and analgesics* – do not modify disease but help relieve symptoms during flares
- *Joint injections, increasingly under ultrasound guidance* – effective, first-line treatment for oligoarticular JIA; in polyarticular disease multiple joint injections are used as bridging agent when starting methotrexate. Often requires sedation or inhaled anaesthesia (Entonox)
- *Methotrexate* – early use reduces joint damage. Effective in approximately 70% with polyarthritis, less effective in systemic features of JIA. It is given as weekly dose (tablet, liquid or injection) and regular blood monitoring is required (for abnormal liver function and bone-marrow suppression). Nausea is common.

- *Systemic corticosteroids* – avoided if possible, to minimise risk of growth suppression and osteoporosis. Pulsed intravenous methylprednisolone often used for severe polyarthritis as an induction agent. May be life-saving for severe systemic arthritis or macrophage activation syndrome.
- *Cytokine modulators* (*'biologics'*) *and other immunotherapies* – Many agents (e.g. anti-TNF alpha, IL-1, CTLA-4 or IL-6) now available and useful in severe disease refractory to methotrexate. Costly and given under strict national guidance with registries for long-term surveillance. T-cell depletion coupled with autologous haematopoetic stem cell rescue (bone marrow transplant) is an option for refractory disease.

Prognosis

Long-term outcome studies have shown that at least one in three children will have ongoing active disease into adult years, with significant morbidity from previous inflammation, such as joint damage requiring joint replacement surgery, visual impairment from uveitis, or fractures from osteoporosis. There is also significant psychosocial morbidity. However, with current management approaches it is anticipated that long-term outcomes will improve.

Transitional care programmes are increasingly provided to facilitate the changes through adolescence and young adulthood and to help young people learn how to manage their chronic disease independently.

Henoch–Schönlein purpura

This is the most common vasculitis of childhood and presents with a purpuric rash over the lower legs and buttocks, usually associated with arthritis of the ankles or knees. Other features are abdominal pain, haematuria and proteinuria (see Ch. 18).

Systemic lupus erythematosus

Systemic lupus erythematosus is rare in children, but may present in adolescent females typically with

Summary

Diagnostic clues regarding musculoskeletal disorders

'Typical' symptom combinations	Pivotal clinical features	Possible diagnoses
Nocturnal wakening with leg pain	Normal child Anaemia, bruising, irritability, infections	'Growing pains' Osteoid osteoma Leukaemia, lymphoma, neuroblastoma (young child)
'Clunk' on hip movement on screening, limp in an older infant	Asymmetrical upper leg skin folds, limited hip abduction	Developmental dysplasia of the hip (DDH)
Febrile, toxic-looking infant, irritability with nappy changing	Restricted joint range (especially hip)	Septic arthritis Osteomyelitis
Sudden limp in a otherwise well young child	Unilateral restricted hip movement	Transient synovitis of the hip Perthes disease
Fever, erythematous rash, red eyes, irritability in infant or young child	Erythema/oedema of hands and feet, oral mucositis, cervical lymphadenopathy	Kawasaki disease
Irritability, fever, reluctance to move in an infant or young child	Stiff back, 'tripod' sitting	Discitis Vertebral osteomyelitis
Joint pain, stiffness and restriction **Loss of joint function**	Persistent joint swelling Loss of joint range	Juvenile idiopathic arthritis
Hip pain in an obese adolescent boy	Unilateral hip restriction	Slipped capital femoral epiphysis
Lethargy, unwilling to do physical activities, irritability, rash	Eyelid erythema Proximal muscle weakness	Juvenile dermatomyositis
Constitutional symptoms, lethargy, arthralgia in an adolescent female	Multi-system abnormalities, haematuria, facial erythema	Systemic lupus erythematosus

malaise, arthralgia and malar rash (often photosensitive). Organ involvement (kidneys, lung or CNS) are serious complications.

Juvenile dermatomyositis

Juvenile dermatomyositis (JDMS) is rare. It usually begins insidiously with malaise, progressive weakness (often difficulty climbing stairs) and facial rash with erythema over the bridge of the nose and malar areas and a violaceous (heliotropic) discoloration of the eyelids (see Fig. 27.10). The skin over the metacarpal and proximal interphalangeal joints may be hypertrophic and pink, and the nailfold capillaries may be dilated and tortuous. Muscle pain is a common, if non-specific, symptom and arthritis is present in 30%. Respiratory failure and aspiration pneumonia may be life-threatening. The condition is described further in Chapter 27.

Genetic skeletal conditions

These are inherited abnormalities resulting in generalised developmental disorders of the bone, of which there are several hundred types. They usually result in reduced growth and abnormality of bone shape rather than impaired strength, except for osteogenesis imperfecta. The bones of the limbs and spine are often affected, resulting in short stature. Intelligence is usually normal. Improved knowledge of the molecular basis of collagen and its disorders is allowing better understanding and delineation of some of these disorders.

Achondroplasia

Inheritance is autosomal dominant, but about 50% are new mutations. Clinical features are short stature from marked shortening of the limbs, a large head, frontal bossing and depression of the nasal bridge (see Fig. 11.10). The hands are short and broad. A marked lumbar lordosis develops. Hydrocephalus sometimes occurs.

Thanatophoric dysplasia

This results in stillbirth. The infants have a large head, extremely short limbs and a small chest. The appearance of the bones on X-ray is characteristic. The importance of the correct diagnosis of this disorder is that, in contrast to achondroplasia, its inheritance is sporadic. It may be identified on antenatal ultrasound.

Cleidocranial dysostosis

In this autosomal dominant disorder, there is absence of part or all of the clavicles and delay in closure of the anterior fontanelle and of ossification of the skull. The child is often able to bring the shoulders together in front of the chest to touch each other as a 'party trick'. Short stature is usually present. Intelligence is normal.

Arthrogryposis

This is a heterogeneous group of congenital disorders in which there is stiffness and contracture of joints. The cause is usually unknown, but there may be an association with oligohydramnios, widespread congenital anomalies or chromosomal disorders. It is usually sporadic. Marked flexion contractures of the knees, elbows and wrists, dislocation of the hips and other joints, talipes equinovarus and scoliosis are common, but the disorder may be localised to the upper or lower limbs. The skin is thin, subcutaneous tissue is reduced and there is marked muscle atrophy around the affected joints. Intelligence is usually unaffected. Management is with physiotherapy and correction of deformities, where possible, by splints, plaster casts or surgery. Walking is impaired in the more severe forms of the disorder.

Osteogenesis imperfecta (brittle bone disease)

This is a group of disorders of collagen metabolism causing bone fragility, with bowing and frequent fractures.

In the most common form (type I), which is autosomal dominant, there are fractures during childhood (Fig. 26.19a) and a blue appearance to the sclerae (Fig. 26.19b) and some develop hearing loss. Treatment with bisphosphonates reduces fracture rates. The prognosis is variable. Fractures require splinting to minimise joint deformity.

There is a severe, lethal form (type II) with multiple fractures already present before birth (Fig. 26.20). Many affected infants are stillborn. Inheritance is variable but mostly autosomal dominant or due to new mutations. In other types, scleral discoloration may be minimal.

Osteopetrosis (marble bone disease)

In this rare disorder, the bones are dense but brittle. The severe autosomal recessive disorder presents with failure to thrive, recurrent infection, hypocalcaemia, anaemia and thrombocytopenia. Prognosis is poor, but bone marrow transplantation can be curative. A less severe autosomal dominant form may present during childhood with fractures.

Marfan syndrome

This is an autosomal dominant disorder of connective tissue associated with tall stature, long thin digits (arachnodactyly), hyperextensible joints, a high arched palate, dislocation (usually upwards) of the lenses of the eyes and severe myopia. The body proportions

Osteogenesis imperfecta



(a)



(b)



Figure 26.20 Osteogenesis imperfecta (type II) showing shortened, deformed lower limbs from gross deformity of the bones with multiple fractures.

Figure 26.19 Osteogenesis imperfecta type I, showing **(a)** fracture of the humerus and osteoporotic bones, **(b)** blue sclerae.

Osteogenesis imperfecta is often considered in the evaluation of unexplained fractures in suspected child abuse.

are altered, with long, thin limbs resulting in a greater distance between the pubis and soles (lower segment) than from the crown to the pubis (upper segment). The arm span, measured from the extended fingers, is greater than the height. There may be chest deformity and scoliosis. The major problems are cardiovascular, due to degeneration of the media of vessel walls resulting in a dilated, incompetent aortic root with valvular incompetence and mitral valve prolapse and regurgitation. Aneurysms of the aorta may dissect or rupture. Monitoring by echocardiography is required.

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27

Neurological disorders

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The central nervous system comprises 100 000 million neurones and when it malfunctions it has the potential to generate a wide spectrum of clinical problems. The site of the dysfunctional neurones determines the nature of the problem, which may involve impaired movement, vision, hearing, sensory perception, memory or consciousness. Classifying this wide range of symptom complexes can be problematic.

Headache

Headache is a frequent reason for older children and adolescents to consult a doctor. The International Headache Society (IHS) has devised a classification, as shown in Box 27.1, which defines:

- Primary headaches: four main groups, comprising migraine, tension-type headache, cluster headache (and other trigeminal autonomic cephalalgias); and other primary headaches (such as cough or exertional headache). They are thought to be due to a primary malfunction of neurones.
- Secondary headaches: symptomatic of some underlying pathology, e.g. from raised intracranial pressure and space-occupying lesions
- Trigeminal and other cranial neuralgias, and other headaches including root pain from herpes zoster.

Primary headaches

Tension-type headache

This is a symmetrical headache of gradual onset, often described as tightness, a band or pressure. There are usually no other symptoms.

Migraine without aura

This accounts for 90% of migraine. In children, episodes may last 1–72 h; the headache is commonly bilateral but may be unilateral. Characteristically pulsatile, over temporal or frontal area, it is often accompanied by unpleasant gastrointestinal disturbance such as nausea, vomiting and abdominal pain and photophobia or phonophobia (sensitivity to sounds). Aggravated by physical activity.

Migraine with aura

Accounts for 10% of migraine. The headache is preceded by an aura (visual, sensory or motor), although the aura may occur without a headache. Features are the absence of problems between episodes and the frequent presence of premonitory symptoms (tiredness, difficulty concentrating, autonomic features, etc.).

The most common aura comprises visual disturbance, which may include:

- Negative phenomena, such as hemianopia (loss of half the visual field) or scotoma (small areas of visual loss)
- Positive phenomena such as fortification spectra (seeing zigzag lines).

Rarely, there are unilateral sensory or motor symptoms.

Episodes usually last for a few hours, during which time children often prefer to lie down in a quiet, dark place. Sleep often relieves the bout.

Symptoms of *tension-type headache* or a *migraine* often overlap. They are probably part of the same pathophysiological continuum, with evidence that both result from primary neuronal dysfunction,

Box 27.1 The classification of headache disorders

Primary headaches
- Migraine
- Tension-type headache
- Cluster headache and other trigeminal autonomic cephalalgias
- Other primary headaches

Secondary headaches

Headache attributed to:
- Head and/or neck trauma
- Cranial or cervical vascular disorder – vascular malformation or intracranial haemorrhage
- Non-vascular intracranial disorder – raised intracranial pressure, idiopathic intracranial hypertension
- A substance or its withdrawal – alcohol, solvent or drug abuse
- Infection – meningitis or encephalitis
- Disorder of homeostasis – hypercapnia or hypertension
- Disorder of facial or cranial structures – acute sinusitis
- Psychiatric disorder

Cranial neuralgias, central and primary facial pain and other headaches
- Trigeminal and other cranial neuralgias and central causes of facial pain
- Other headaches.

including channelopathies, with vascular phenomena as secondary events. There is a genetic predisposition, with first- and second-degree relatives often also affected. Bouts are often triggered by a disturbance of inherent biorhythms, such as late nights or early rises, stress, or winding down after stress at home or school. Certain foods, e.g. cheese, chocolate and caffeine, are only rarely a reliable trigger. In girls, headaches can be related to menstruation and the oral contraceptive pill.

Uncommon forms of migraine

These include:

- *Familial* – linked to a calcium channel defect, dominantly inherited
- *Sporadic hemiplegic migraine*
- *Basilar-type migraine* – vomiting with nystagmus and/or cerebellar signs
- *Periodic syndromes* – often precursors of migraine and include:
  - *Cyclical vomiting* – recurrent stereotyped episodes of vomiting and intense nausea associated with pallor and lethargy. The child is well in between.
  - *Abdominal migraine* – an idiopathic recurrent disorder characterised by episodic midline abdominal pain in bouts lasting 1–72 h. Pain is moderate to severe in intensity and associated with vasomotor symptoms, nausea and vomiting. The child is well in between episodes.
  - *Benign paroxysmal vertigo* of childhood – a heterogeneous disorder characterised by recurrent brief episodes of vertigo occurring without warning and resolving spontaneously in otherwise healthy children. Between episodes, neurological examination, audiometric and vestibular function tests are normal.

Secondary headaches

Raised intracranial pressure and space-occupying lesions

Headaches often raise the fear of brain tumours; it may well be the reason for parents to consult a doctor. Headaches due to a space-occupying lesion are worse when lying down and morning vomiting is characteristic. The headaches may also cause night-time waking. There is often a change in mood, personality or educational performance. Other features suggestive of a space-occupying lesion are:

- Visual field defects – from lesions pressing on the optic pathways, e.g. craniopharyngioma (a pituitary tumour)
- Cranial nerve abnormalities causing diplopia, new-onset squint or facial nerve palsy. The VIth (abducens) cranial nerve has a long intracranial course and is often affected when there is raised pressure, resulting in a squint with diplopia and inability to abduct the eye beyond the midline. It is a false localising sign. Other nerves are affected depending on the site of lesion, e.g. pontine lesions may affect the VIIth (facial) cranial nerve and cause a facial nerve palsy
- Abnormal gait
- Torticollis (tilting of the head)
- Growth failure, e.g. craniopharyngioma or hypothalamic lesion
- Papilloedema – a late feature
- Cranial bruits – may be heard in arteriovenous malformations but these lesions are rare.

Other causes

These are listed in Box 27.1.

Management

The mainstay of management is a thorough history and examination with detailed explanation and advice. Imaging is unnecessary in the absence of any Red Flag features.

Children and parents should be informed that recurrent headaches are common. For most there are good and bad spells, with periods of months or even years in between the bad spells, and that they cause no long-term harm. Written child-friendly information for the family to take home is helpful. Children should be advised on how to live with and control the headaches, rather than allowing the headaches to dominate their lives. There is nothing medicine can do to cure this problem but there is much it can offer to make the bad spells more bearable.

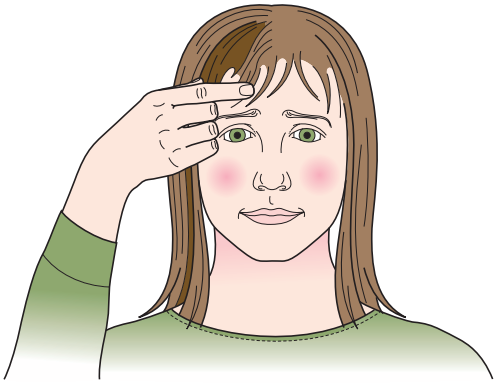
Rescue treatments

- Analgesia – paracetamol and non-steroidal anti-inflammatory drugs (NSAIDs), taken as early as possible in an individual troublesome episode
- Anti-emetics – prochlorperazine and metoclopramide
- Serotonin (5-HT_1) agonists, e.g. sumatriptan. A nasal preparation of this is licensed for use in children over 12 years of age.

Summary

History

Premonitory symptoms, aura, character, position, radiation, frequency, duration, triggers, relieving and exacerbating factors?
Special consideration:
Triggers – stress, relaxation, food, menstruation?
Emotional or behavioural problems at home or school?
Vision checked – refractive error?
Head trauma?
Alcohol, solvent, or drug abuse?
Analgesia over-use?



Investigations

Only consider these if Red Flag features

Headache type

Tension-type headache – constriction band.
Migraine without aura – bilateral or unilateral, pulsatile, gastrointestinal disturbance, e.g. nausea, vomiting, abdominal pain, photophobia. Lies in quiet, dark place. Relieved by sleep
Migraine with aura – preceded by aura (visual, sensory or motor), premonitory symptoms
Mixed-type headaches – common

Red flag symptoms – space-occupying lesion

Headache – worse lying down or with coughing and straining
Headache – wakes up child (different from headache on awakening, not uncommon in migraine)
Associated confusion, and/or morning or persistent nausea or vomiting
Recent change in personality, behaviour or educational performance

Red flag physical signs – space-occupying lesion

- Growth failure
- Visual field defects – craniopharyngioma
- Squint
- Cranial nerve abnormality
- Torticollis
- Abnormal coordination – for cerebellar lesions
- Gait – upper motor neurone or cerebellar signs
- Fundi – papilloedema
- Bradycardia
- Cranial bruits – arteriovenous malformation

Other physical signs

Visual acuity – for refractive errors
Sinus tenderness – for sinusitis
Pain on chewing – temporomandibular joint malocclusion
Blood pressure – for hypertension

Prophylactic agents

Where headaches are frequent and intrusive:

- Pizotifen (5-HT antagonist) – can cause weight gain and sleepiness
- Beta-blockers – propranolol; contraindicated in asthma.
- Sodium channel blockers – valproate or topiramate.

Psychosocial support

- Psychological support – is it required to ameliorate a particular stressor, e.g. bullying, anxiety over exams or illness in friends or family?
- Relaxation and other self-regulating techniques.

Systematic reviews highlight the need for further evidence on which to base headache management.

Seizures

A seizure is a clinical event in which there is a sudden disturbance of neurological function caused by an abnormal or excessive neuronal discharge. Seizures may be epileptic or non-epileptic.

The causes of seizures are listed in Box 27.2.

Febrile seizures

A febrile seizure is a seizure accompanied by a fever in the absence of intracranial infection due to bacterial meningitis or viral encephalitis. These occur in 3% of children, between the ages of 6 months and 5 years. There is a genetic predisposition, with a 10% risk if the child has a first-degree relative with febrile seizures. The seizure usually occurs early in a viral infection when the temperature is rising rapidly. The seizures are usually brief, and are generalised tonic-clonic seizures. About 30–40% will have further febrile seizures. This is more likely the younger the child, the shorter the duration of illness before the seizure, the lower the temperature at the time of seizure and if there is a positive family history.

Simple febrile seizures do not cause brain damage; the child's subsequent intellectual performance is the same as in children who do not experience a febrile seizure. There is a 1–2% chance of developing epilepsy, similar to the risk for all children.

However, complex febrile seizures; i.e. those which are focal, prolonged, or repeated in the same illness, have an increased risk of 4–12% of subsequent epilepsy.

The acute management of seizures is described in Chapter 6. Examination should focus on the cause of the fever, which is usually a viral illness, but a bacterial infection including meningitis should always be considered. The classical features of meningitis such as neck stiffness and photophobia may not be as apparent in children <18 months of age, so an infection screen (including blood cultures, urine culture and lumbar puncture for CSF) may be necessary. If the child is unconscious or has cardiovascular instability, lumbar puncture is contraindicated and antibiotics should be started immediately.

Parents need reassurance and information. Advice sheets are usually given to parents. Antipyretics have not been shown to prevent febrile seizures and tepid sponging is no longer recommended. The family should be taught the first aid management of seizures. If there is a history of prolonged seizures (>5 min), rescue therapy with rectal diazepam or buccal midazolam can be supplied. Oral prophylactic anti-epileptic drugs are not used as they do not reduce the recurrence rate of seizures or the risk of epilepsy. An EEG is not indicated as it does not serve as a guide for treatment; nor does it predict seizure recurrence.

Box 27.2 Causes of seizures

Epilepsy

- Idiopathic (70–80%) – cause unknown but presumed genetic
- Secondary
  - Cerebral dysgenesis/malformation
  - Cerebral vascular occlusion
  - Cerebral damage, e.g. congenital infection, hypoxic-ischaemic encephalopathy, intraventricular haemorrhage/ischaemia
- Cerebral tumour
- Neurodegenerative disorders
- Neurocutaneous syndromes

Non-epileptic

- Febrile seizures
- Metabolic
  - Hypoglycaemia
  - Hypocalcaemia/hypomagnesaemia
  - Hypo/hypernatraemia
- Head trauma
- Meningitis/encephalitis
- Poisons/toxins.

Summary

Febrile seizures

- Affect 3% of children; have a genetic predisposition
- Occur between 6 months and 6 years of age
- Are usually brief, generalised tonic-clonic seizures occurring with a rapid rise in fever
- If a bacterial infection, especially meningitis, is present, it needs to be identified and treated
- Advise family about management of seizures, consider rescue therapy
- If simple – does not affect intellectual performance or risk of developing epilepsy
- If complex, 4–12% risk of subsequent epilepsy.

Paroxysmal disorders

There is a broad differential diagnosis for children with paroxysmal disorders ('funny turns'). Epilepsy is a clinical diagnosis based on the history from eyewitnesses and the child's own account. If available, videos of the seizures or suspected seizures can be of great help. The diagnostic question is whether the paroxysmal events are that of an epilepsy of childhood or one of the many conditions which mimic it (Fig. 27.1). The most common pitfall is that of syncope leading to an anoxic (non-epileptic) tonic-clonic seizure.

The key to the diagnosis lies in a detailed history, which, together with clinical examination, will determine the need for an EEG or other investigations.

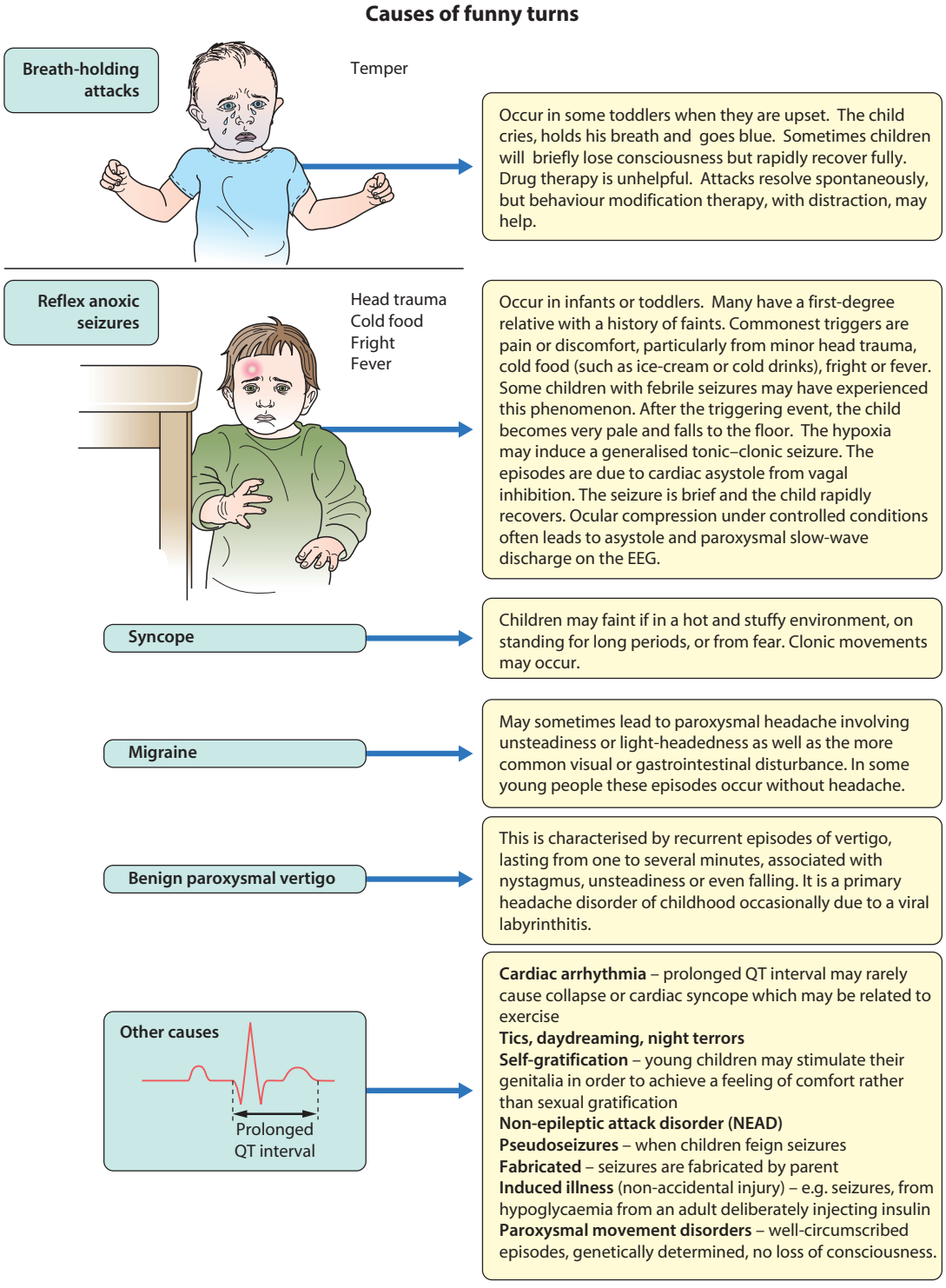


Figure 27.1 Causes of 'funny turns'.

Summary

Breath-holding and reflex anoxic seizures

In toddlers:

- Breath-holding episodes – toddler, precipitated by anger, holds breath, goes blue, then limp, rapid recovery
- Reflex anoxic seizures – toddler, precipitated by pain, stops breathing, goes pale, brief seizure sometimes, rapid recovery
- Other non-epileptic paroxysmal disorders: see Fig. 27.1.

Epilepsies of childhood

Epilepsy has an incidence of about 0.05% (after the first year of life when it is even more common) and a prevalence of 0.5%. This means that most large secondary schools will have about six children with an epilepsy. Epilepsy is a chronic neurological disorder characterised by recurrent unprovoked seizures, consisting of transient signs and/or symptoms associated with abnormal, excessive or synchronous neuronal activity in the brain. Most epilepsy is idiopathic but other causes of seizures are listed in Box 27.2.

An international classification of epilepsy is used, which has recently been revised (International League Against Epilepsy (ILAE) 2010 Classifications). This broadly classifies seizures as either:

- *Generalised* – discharge arises from both hemispheres. They may be – absence, myoclonic, tonic, tonic-clonic and atonic
- *Focal* – where seizures arise from one or part of one hemisphere.

Focal seizure manifestations will depend on the part of the brain where the discharge originates:

- *Frontal* seizures – involve the motor or premotor cortex. May lead to clonic movements, which may travel proximally (Jacksonian march). Asymmetrical tonic seizures can be seen, which may be bizarre and hyperkinetic and can be mistakenly dismissed as non-epileptic events. Atonic seizures may arise from mesial frontal discharge.
- *Temporal lobe* seizures, the most common of all the epilepsies – may result in strange warning feelings or aura with smell and taste abnormalities and distortions of sound and shape. Lip-smacking, plucking at one's clothing and walking in a non-purposeful manner (automatisms) may be seen, following spread to the pre-motor cortex. Déjà-vu and jamais-vu are described (intense feelings of having been, or never having been, in the same situation before). Consciousness can be impaired and the length of event is longer than a typical absence.
- *Occipital* seizures – cause distortion of vision.
- *Parietal lobe* seizures – cause contralateral dysaesthesias (altered sensation), or distorted body image.

In focal seizures, the level of consciousness may be retained, consciousness may be lost, or the seizure may be followed by generalised tonic-clonic seizure. In the new classification, the terms, simple or complex or discognitive are no longer used and the impairment of consciousness is not classified but described.

In many children, especially under 5 years old, it may be unclear whether a seizure is generalised or focal.

The main seizure types are summarised in Figure 27.2 and the epilepsy syndromes in Table 27.1.

Diagnosis

The diagnosis of epilepsy is primarily based on a detailed history from the child and eyewitnesses, substantiated by a video if available. This is increasingly available on mobile phones. Particular attention is focussed on any specific triggers and if the child has any impairments, as there may be educational, psychological or social problems. Clinical examination should include checking for skin markers for a neurocutaneous syndrome or neurological abnormalities. Although epilepsy is usually idiopathic, it may be the presentation or a complication of an underlying neurological disorder.

Investigation of seizures

EEG

An EEG is indicated whenever epilepsy is suspected. It is interpreted to identify a background that is abnormal for the child's age; asymmetry or slowing that might suggest underlying structural abnormalities; or evidence of neuronal hyperexcitability such as sharp waves or spike-wave complexes. Many children with epilepsy have a normal initial EEG; and many children who will never have epilepsy have EEG abnormalities. Unless a seizure is actually captured, an EEG does no more than add supportive evidence (or not) for the diagnosis. If the standard EEG is normal, a sleep or sleep-deprived record can be helpful. Additional techniques are 24–h ambulatory EEG or, ideally, video-telemetry. For assessment prior to surgery, more invasive techniques such as subdural electrodes can be used.

Imaging

- *Structural*. MRI and CT brain scans are not required routinely for childhood generalised epilepsies. They are indicated if there are neurological signs between seizures, or if seizures are focal, in order to identify a tumour, vascular lesion, or area of sclerosis which could be treatable. MRI FLAIR (fluid-attenuated inversion recovery) sequences better detect mesial temporal sclerosis in temporal lobe epilepsy.
- *Functional scans*. While it is not always possible to see structural lesions, techniques have advanced to allow functional imaging to detect areas of abnormal metabolism suggestive of seizure foci. These include PET (positron emission tomography) and SPECT (single positron emission computed tomography), which use isotopes and ligands, injected and taken up by metabolically active cells. Both can be used between seizures to detect areas of hypometabolism in epileptogenic lesions. SPECT

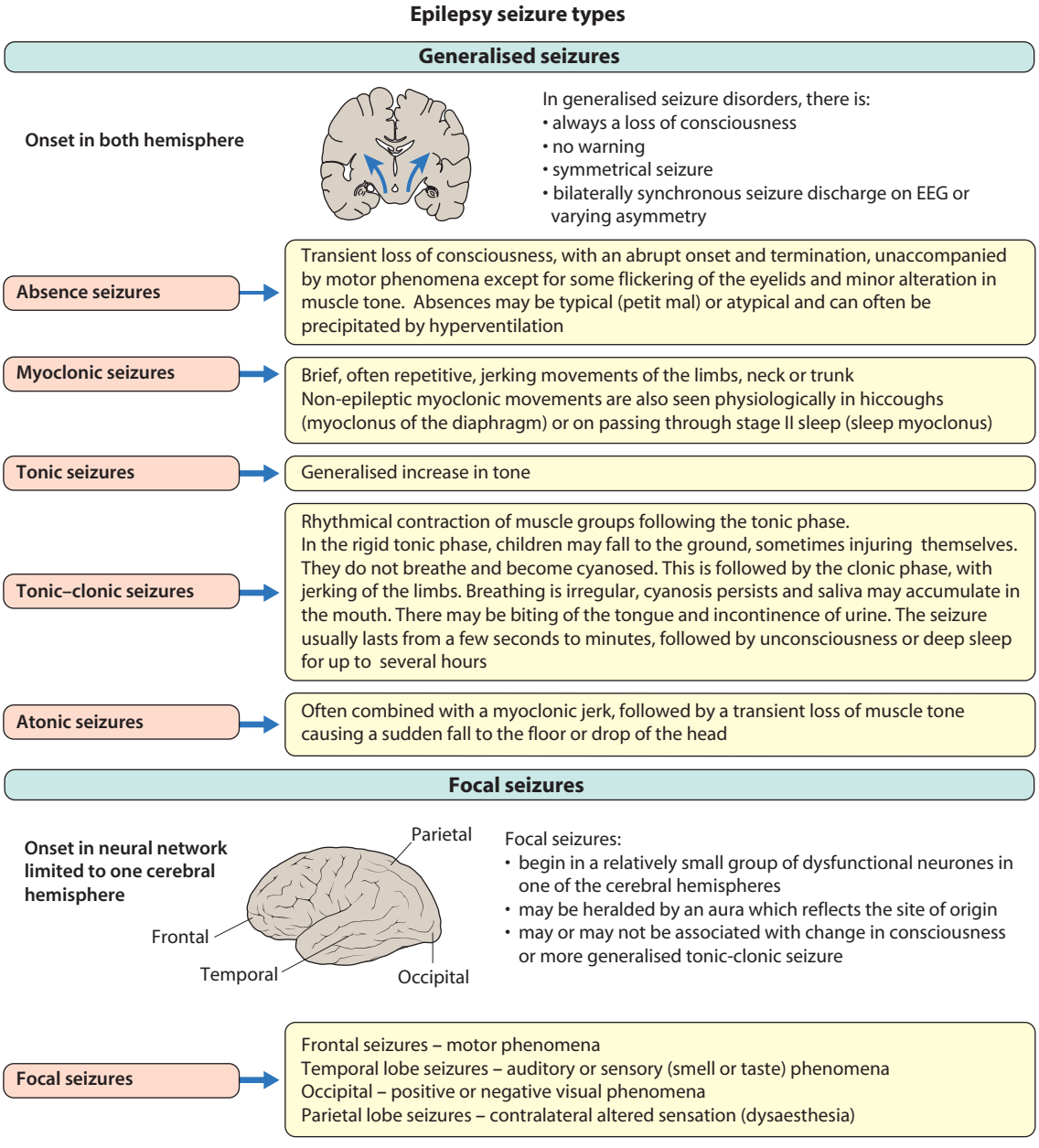


Figure 27.2 Epilepsy seizure types.

can also be used to capture seizures and areas of hypermetabolism. Functional MRI can be used alongside psychological testing –including memory assessment – to minimise the risk of postoperative impairment.

Other investigations

Metabolic investigations may be warranted when there is developmental regression or seizures are related to feeds or fasting. Genetic studies will become increasingly helpful as certain epilepsy syndromes are now known to be due to genetic deletions causing abnormalities of sodium and other ion channels (e.g. SCN1A mutations in severe myoclonic epilepsy of infancy).

Management

Management begins with explanation and advice to help adjustment to the condition. A specialist epilepsy nurse may assist families by providing education and continuing advice on lifestyle issues. The decision whether to treat or not is related to the level of inconvenience seizures are bringing into the young person's life. It is common practice not to institute treatment after a single unprovoked seizure.

Anti-epileptic drug (AED) therapy

Principles governing use are:

- Not all seizures require AED therapy. This decision should be based on the seizure type, frequency

Table 27.1 Some epilepsy syndromes – arranged by age of onset

Name	Age	Seizure pattern	Comments
West syndrome	4–6 months	Violent flexor spasms of the head, trunk and limbs followed by extension of the arms (so-called 'salaam spasms'). Flexor spasms last 1–2 s, often multiple bursts of 20–30 spasms, often on waking, but may occur many times a day. May be misinterpreted as colic. Social interaction often deteriorates – a useful marker in the history	Many causes; two-thirds have underlying neurological cause. The EEG shows hypsarrhythmia, a chaotic pattern of high-voltage slow waves, and multi-focal sharp wave discharges (Fig. 27.3). Treatment is with vigabatrin or corticosteroids; good response in 30–40%, but unwanted effects are common. Most will subsequently lose skills and develop learning disability or epilepsy
Lennox–Gastaut syndrome	1–3 years	Multiple seizure types, but mostly drop attacks (astatic seizures), tonic seizures and atypical absences. Also neurodevelopmental arrest or regression and behaviour disorder	Often other complex neurological problems or history of infantile spasms. Prognosis is poor
Childhood absence epilepsy	4–12 years	Stare momentarily and stop moving, may twitch their eyelids or a hand minimally. Lasts only a few seconds and certainly not longer than 30 s. Child has no recall except realises they have missed something and may look puzzled or say 'pardon' on regaining consciousness. Developmentally normal but can interfere with schooling. Accounts for only 2% of childhood epilepsy	Two-thirds are female. The episodes can be induced by hyperventilation, the child being asked to blow on a piece of paper or windmill for 2–3 min, a useful test in the outpatient clinic. The EEG shows generalised 3/second spike and wave discharge, which is bilaterally synchronous during and sometimes between episodes (Fig. 27.4). Prognosis is good, with 95% remission in adolescence; 5–10% may develop tonic-clonic seizures in adult life
Benign* epilepsy, with centrotemporal spikes (BECTS)	4–10 years	Tonic-clonic seizures in sleep, or simple focal seizures with awareness of abnormal feelings in the tongue and distortion of the face (supplied by the Rolandic area of the brain)	Comprises 15% of all childhood epilepsies. EEG shows focal sharp waves from the Rolandic or centrotemporal area. Important to recognise as it is benign and does not always require treatment. Almost all remit in adolescence
Early-onset benign* childhood occipital epilepsy (Panayiotopoulos type)	1–14 years	Younger children – periods of unresponsiveness, eye deviation, vomiting and autonomic features. Older children – headache and visual disturbance including distortion of images and hallucinations	Uncommon. EEG shows occipital discharges. Remit in childhood
Juvenile myoclonic epilepsy	Adolescence-adulthood	Myoclonic seizures, but generalised tonic-clonic seizures and absences may occur, mostly shortly after waking. A typical history is throwing drinks or cornflakes about in the morning as myoclonus occurs at this time. Learning is unimpaired	Characteristic EEG. Response to treatment is usually good but lifelong. A genetic linkage has been identified. Remission unlikely

*Although called benign, may be specific learning difficulties in some children.

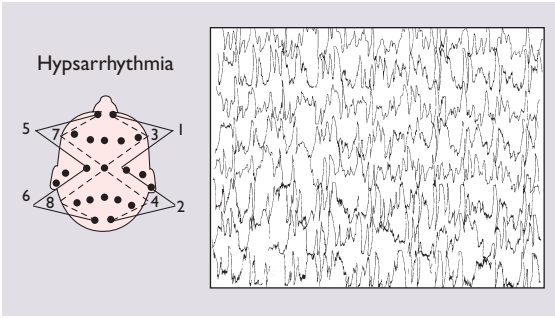


Figure 27.3 EEG of hypsarrhythmia in West syndrome. There is a chaotic background of slow-wave activity with sharp multi-focal components.

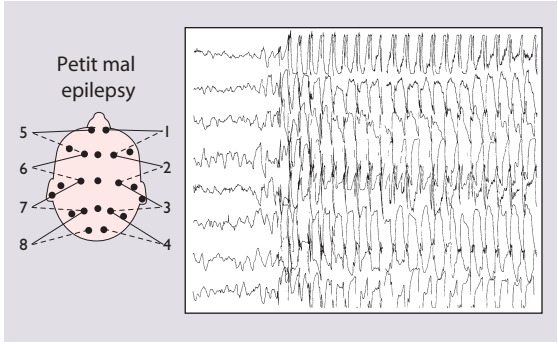


Figure 27.4 EEG in a typical absence seizure in childhood absence epilepsy. There is three per second spike and wave discharge which is bilaterally synchronous during, and sometimes between, attacks.

and the social and educational consequences of the seizures set against the possibility of unwanted effects of the drugs.

- Choose the appropriate drug for the seizure. Inappropriate AEDs may be detrimental, e.g. carbamazepine can make absence and myoclonic seizures worse.
- Monotherapy at the minimum dosage is the desired goal, although in practice more then one drug may be required.
- All AEDs have potential unwanted effects and these should be discussed with the child and parent.
- Drug levels are not measured routinely, but may be useful to check for adherence to advice or with some drugs with erratic pharmacokinetics, e.g. phenytoin.
- Children with prolonged seizures are given rescue therapy to keep with them. This is usually a benzodiazepine, e.g. rectal diazepam or buccal midazolam.
- AED therapy can usually be discontinued after 2 years free of seizures.

Guidance regarding treatment options for different seizure types are shown in Table 27.2. Common unwanted effects of AEDs are shown in Table 27.3.

Other treatment options

In children with intractable seizures, there are a number of radical treatment options.

- *Ketogenic* (*fat-based*) *diets* may be helpful in some children. Its mechanism of action is poorly understood.
- *Vagal nerve stimulation*, delivered using externally programmable stimulation of a wire implanted around the vagal nerve, may possibly be useful; trials are being conducted.
- *Surgery*. Cessation of seizures and drug therapy may be achieved in some children whose clinical seizures are well-localised as demonstrated by good concordance between EEG, MRI and functional imaging findings. The main procedure is temporal lobectomy for mesial temporal sclerosis, but other procedures include hemispherectomy or hemispherotomy (isolation of hemisphere which is not removed so as avoid post-operative shifts in space) and other focal resections. Detailed assessment is required to ensure that the benefits outweigh the risks.

Advice and prognosis

The aim is to promote independence and confidence. Some children with epilepsy and their families need psychological help to adjust to the condition. The school needs to be aware of the child's problem and teachers advised on the management of seizures. Unrecognised absences may interfere with learning, which is an indication for being vigilant about 'odd

Table 27.2 Choice of anti-epileptic drugs (NICE 2004)

Seizure type	First-line	Second-line
Generalised seizures		
Tonic-clonic	Valproate, carbamazepine	Lamotrigine, topiramate
Absence	Valproate, ethosuximide	Lamotrigine
Myoclonic	Valproate	Lamotrigine
Focal seizures	Carbamazepine, valproate Lamotrigine shown since to be most effective – but slow titration	Topiramate, levetiracetam, oxcarbazepine, gabapentin, tiagabine, vigabatrin

Table 27.3 Common or important unwanted effects of anti-epileptic drugs

Drug	Side-effects
Valproate	Weight gain, hair loss
	Rare idiosyncratic liver failure
Carbamazepine/oxcarbazepine	Rash, neutropenia, hyponatraemia, ataxia
	Liver enzyme induction, can interfere with other medication
Vigabatrin	Restriction of visual fields, which has limited its use
	Sedation
Lamotrigine	Rash
Ethosuximide	Nausea and vomiting
Topiramate	Drowsiness, withdrawal and weight loss
Gabapentin	Insomnia
Levetiracetam	Sedation – rare
Benzodiazepines – clobazam, clonazepam, diazepam, nitrazepam	Sedation, tolerance to effect, increased secretions

All the above may cause drowsiness and occasional skin rashes.

episodes' which may represent seizures. Relatively few restrictions are required, but situations where having a seizure could lead to injury or be fatal should be avoided. This includes avoiding deep baths (showers are preferable) and not swimming alone in deep water. Those with photosensitivity should sit at a distance from televisions, can cover one eye, and check that TVs and VDUs in use are acceptable (Epilepsy Action consider most modern TVs and VDUs to be suitable and can provide advice).

For adolescents, there may be issues to discuss around driving (only after 1 year free of seizures), contraception and pregnancy. There may also be issues with adherence and precipitation of seizures by alcohol and poor sleep routines.

Sudden unexpected death in epilepsy, SUDEP, may be discussed, and its low risks emphasised. Information is available from self-help groups and organisations such as Epilepsy Action.

Children with epilepsy do less well educationally, with social outcomes and with future employment than those with other chronic illnesses such as diabetes.

Two-thirds of children with epilepsy go to a mainstream school, but some require educational help for associated learning difficulties. One-third attend a special school, but they often have multiple disabilities and their epilepsy is part of a severe brain disorder. A few children require residential schooling where there are facilities and expertise in monitoring and treating intractable seizures.

Status epilepticus

Status epilepticus, a seizure lasting 30 min or repeated seizures for 30 min without recovery of consciousness is described in Chapter 6.

Summary

Epilepsy

- Affects 1 in 200 children
- Classified according to seizure type; the identification of a syndrome, where possible; and underlying aetiology
- If suspected, an EEG is indicated
- Anti-epileptic drug therapy should be considered where the seizures are intrusive, selected according to seizure type, monotherapy if possible and with the least potential for unwanted effects
- Requires liaison with the school about the management of seizures and avoiding situations which could lead to injury.

Motor disorders

Movement is governed by three main cerebral control centres. Patterns of information, modulated by afferent sensory information (joint position, crude touch, visual, auditory and vestibular), pass down the brainstem and spinal cord, through synapses in the anterior horns and along peripheral nerves to the target muscles. In clinical practice the first question to ask when seeing a child with a motor disorder is whether this is a central or a peripheral nervous system disorder. The pattern of movement usually gives the answer.

Table 27.4 Causes of movement disorders

Corticospinal (pyramidal) tract disorders	Basal ganglia disorders	Cerebellar disorders
Cerebral dysgenesis, e.g. neuronal migration problem	Acquired brain injury: – Acute and profound hypoxia-ischaemia – Carbon monoxide poisoning – Post cardiopulmonary bypass chorea	Acute – medication and drugs, including alcohol and solvent abuse
Global hypoxia-ischaemia	Post-streptococcal chorea (rheumatic fever)	Post-viral – particularly varicella infection
Arterial ischaemic stroke	Mitochondrial cytopathies	Posterior fossa lesions or tumours, e.g. medulloblastoma
Cerebral tumour	Wilson disease	Genetic and degenerative disorders, e.g. ataxic cerebral palsy, Friedreich ataxia and ataxia-telangiectasia
Acute disseminated encephalomyelitis	Huntington disease	
Post-ictal paresis		
Hemiplegic migraine		

Central motor disorders

The three central movement control centres are:

- *Motor cortex*, lying along the pre-central gyrus (the homunculus reflects the body upside down, legs superiorly and face inferiorly, just above the Sylvian fissure; with large areas to govern fine movements of the tongue, fingers and thumb). Information from here passes down the corticospinal (pyramidal) tracts to link with the basal ganglia.
- *Basal ganglia*, deep grey matter structures, store patterns of movement so that we need not put conscious effort into every movement we make.
- *Cerebellum*, acting as the 'air-traffic controller'; receiving feedback on joint position nanosecond by nanosecond to keep us on course as we execute movements.

Disorders of these central movement control centres are:

- *Corticospinal (pyramidal) tract disorders* – there is weakness with a pattern of adduction at the shoulder, flexion at the elbow and pronation of the forearm; adduction and internal rotation at the hip, flexion at hip, knee and plantar flexion at the ankle with brisk hyper-reflexia and extensor plantars. Fine finger movement will be lost.
- *Basal ganglia disorders* – will lead either to difficulty initiating movement, with fluctuating (largely increased) tone – a 'dystonia', or a 'dyskinesia', where packets of movement information are released to give jerky movement (chorea) or writhing movement (athetosis).
- *Cerebellar disorders* – will lead to difficulty holding a posture (particularly with eyes closed); past-pointing (dysmetria); poor alternating movements (dysdiadochokinesis); and a characteristic scanning dysarthria. Posterior sensory pathway problems may give a similar clinical picture but are much rarer in childhood. The gait is wide-based. Associated nystagmus may be seen. Causes of these disorders are listed in Table 27.4.

Cerebral palsy

This is described in Chapter 4.

Peripheral motor disorders: the neuromuscular disorders

Any part of the lower motor pathway can be affected in a neuromuscular disorder, so that anterior horn cell disorders, peripheral neuropathies, disorders of neuromuscular transmission and primary muscle diseases can all occur. The causes of neuromuscular disorders are shown in Figure 27.5. The key clinical feature of a neuromuscular disorder is weakness, which may be progressive or static. Affected children may present with:

- Floppiness
- Delayed motor milestones
- Muscle weakness
- Unsteady/abnormal gait
- Fatiguability
- Muscle cramps (suggesting a metabolic myopathy).

History and examination may provide useful clues. Children with myopathy often show a waddling gait or positive Gowers sign suggestive of proximal muscle weakness. Gowers sign is the need to turn prone to rise to a standing from a supine position. This is normal until the age of 3 years. It is only when children have become very weak that they 'climb up the legs with the hands' to gain the standing position (Fig. 27.6). A pattern of more distal wasting and weakness, particularly in the presence of pes cavus, suggests an

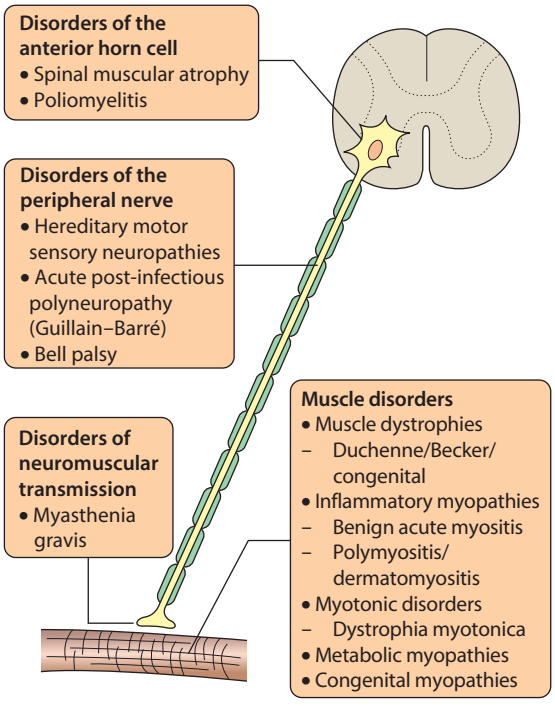
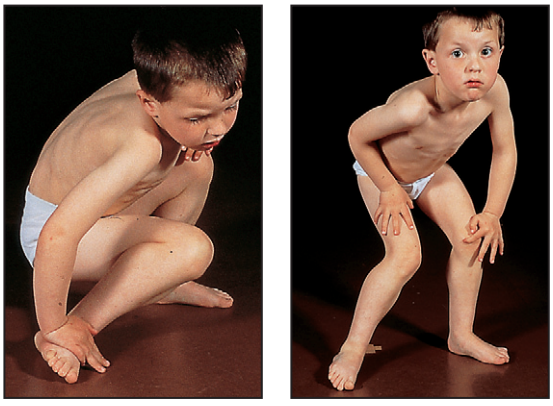


Figure 27.5 Neuromuscular disorders.



(a) (b)

Figure 27.6 (a,b) Gowers sign. The child needs to turn prone to rise (the key, early feature of Gowers sign), then uses his hands to climb up on his knees before standing (late feature), because of poor hip girdle fixation and/or proximal muscle weakness. Any child continuing to turn prone to rise after 3 years of age is likely to have a neuromuscular condition.

hereditary motor sensory neuropathy. Increasing fatiguability through the day, often with ophthalmoplegia and ptosis, would be more consistent with depletion at the motor end-plate and a diagnosis of myasthenia gravis.

It is usually difficult to differentiate a myopathy from a neuropathy on clinical grounds but there are some broad points to look for:

- Anterior horn cell – there are signs of denervation: weakness, loss of reflexes, fasciculation and wasting as the nerve supply to the muscle fails.
- Neuropathy – often distal nerves affected. Motor neuropathy will give weakness, sensory neuropathy will give impaired perception of pain and temperature, or touch, with a loss of reflexes in either.
- Myopathy – there is weakness (often proximal), wasting, gait disturbance.
- Neuromuscular junction – as end-plate acetylcholine stores become depleted, there is diurnal worsening through the day, leading to fatiguability.

Investigations

Myopathy:

- Serum creatine phosphokinase – markedly elevated in Duchenne and Becker muscular dystrophy and inflammatory myopathies
- Muscle biopsy, needle or open – modern histochemical techniques often enable a definitive diagnosis
- DNA testing – to identify abnormal genes
- Ultrasound and MRI of muscles – used in specialist centres to diagnose and monitor progress.

Neuropathy:

- Nerve conduction studies – to identify delayed motor and sensory nerve conduction velocities seen in neuropathy
- DNA testing – for abnormal genes
- Nerve biopsy – rarely performed
- EMG (electromyography) helps in differentiating myopathic from neuropathic disorders, e.g. fatiguability on repetitive nerve stimulation in myasthenia. However, it should be used selectively in children, as the nerve conduction studies cause a tingling sensation and electromyography requires insertion of fine needle electrodes.

Diagnosis of neuromuscular disorders has been made easier by the advances made in confirmatory DNA tests for many of them, e.g. spinal muscular atrophy (SMA), Duchenne muscular dystrophy, myotonic dystrophy, the congenital muscular dystrophies and hereditary neuropathies. This also allows antenatal testing and genetic counselling and often obviates the need for the discomfort of peripheral neurophysiology.

Disorders of the anterior horn cell

Presentation is with weakness, wasting and absent reflexes. The features of poliomyelitis are described in Chapter 14.

Spinal muscular atrophy

This is an autosomal recessive degeneration of the anterior horn cells, leading to progressive weakness and wasting of skeletal muscles due to mutations in the survival motor neurone (SMN) gene. This is the second most common cause of neuromuscular disease in the UK after Duchenne muscular dystrophy. A number of phenotypes are recognised.



Figure 27.7 Spinal muscular atrophy type 1 (Werdnig–Hoffmann disease) showing proximal muscle wasting, chest deformity from weakness of the intercostal muscles and thighs held abducted because of hypotonia.



Figure 27.8 Bell palsy. There is right facial weakness of both the upper and lower face.

Spinal muscular atrophy type 1 (Werdnig–Hoffmann disease)

A very severe progressive disorder presenting in early infancy (Fig. 27.7). Diminished fetal movements are often noticed during pregnancy and there may be arthrogryposis (positional deformities of the limbs with contractures of at least two joints) at birth. Typical signs include:

- Lack of antigravity power in hip flexors
- Absent deep tendon reflexes
- Intercostal recession
- Fasciculation of the tongue.

These children never sit unaided. Death is from respiratory failure within about 12 months. There are milder forms of the disorder with a later onset. Children with type 2 spinal muscular atrophy can sit, but never walk independently. Those with type 3 (Kugelberg–Welander) do walk and can present later in life.

Peripheral neuropathies

The hereditary motor sensory neuropathies (HMSN)

This group of disorders typically leads to symmetrical, slowly progressive muscular wasting, which is distal rather than proximal. Type I, formerly known as peroneal muscular atrophy (Charcot–Marie–Tooth disease), is usually dominantly inherited and the most common. Affected nerves may be hypertrophic due to demyelination followed by attempts at remyelination. Nerve biopsy typically shows 'onion bulb formation' due to these two processes. Onset is in the first decade with distal atrophy and pes cavus, the legs being affected more than the arms. Rarely, there may be distal sensory loss and the reflexes are diminished. The disease is chronic and only rarely do those affected lose the ability to walk. The initial presentation of Friedreich ataxia can be similar.

Acute post-infectious polyneuropathy (Guillain–Barré syndrome)

Presentation is typically 2–3 weeks after an upper respiratory tract infection or campylobacter gastroenteritis. There may be fleeting abnormal sensory symptoms in the legs, but the prominent feature is an ascending symmetrical weakness with loss of reflexes and autonomic involvement. Sensory symptoms, usually in the distal limbs, are less striking than the paresis but can be unpleasant. Involvement of bulbar muscles leads to difficulty with chewing and swallowing and the risk of aspiration. Respiratory depression may require artificial ventilation. The maximum muscle weakness may occur only 2–4 weeks after the onset of illness. Although full recovery may be expected in 95% of cases, this may take up to 2 years.

The CSF protein is characteristically markedly raised, but this may not be seen until the second week of illness. The CSF white cell count is not raised. Nerve conduction velocities are reduced.

Management of post-infectious polyneuropathy is supportive, particularly of respiration. The disorder is probably due to the formation of antibody attaching itself to protein components of myelin. Corticosteroids have no beneficial effect and may delay recovery. Controlled trials indicate the ventilator-dependent period can be significantly reduced by immunoglobulin infusion. If this is not successful, plasma exchange may be effective.

Bell palsy and facial nerve palsies

Bell palsy is an isolated lower motor neurone paresis of the VIIth cranial nerve leading to facial weakness (Fig. 27.8). Although the aetiology is unclear, it is probably post-infectious with an association with herpes simplex virus in adults. Corticosteroids may be of value in reducing oedema in the facial canal during the first week; no benefit from aciclovir has been

demonstrated. Recovery is complete in the majority of cases but may take several months. The main complication is conjunctival infection due to incomplete eye closure on blinking. This may require the eye to be protected with a patch or even tarsorrhaphy.

There is an important differential diagnosis. If symptoms of an VIIIth nerve paresis are also present then the most likely diagnosis is a compressive lesion in the cerebellopontine angle. The herpes virus may invade the geniculate ganglion and give painful vesicles on the tonsillar fauces and external ear, along with a facial nerve paresis. Treatment for this is with aciclovir.

Hypertension should be excluded, as there is an association between Bell palsy and coarctation of the aorta. If the facial weakness is bilateral, sarcoidosis should be suspected, and this is also seen in Lyme disease.

Disorders of neuromuscular transmission

Myasthenia gravis

This presents as abnormal muscle fatiguability which improves with rest or anticholinesterase drugs.

Transient neonatal myasthenia

This is described in Chapter 9.

Juvenile myasthenia

This is similar to adult autoimmune myasthenia and is due to binding of antibody to acetylcholine receptors on the post-junctional synaptic membrane. This gives a reduction of the number of functional receptors. Presentation is usually after 10 years of age with ophthalmoplegia and ptosis, loss of facial expression and difficulty chewing (Fig. 27.9). Generalised, especially proximal, weakness may be seen.

Diagnosis is made by observing improvement following the administration of intravenous edrophonium and can be further confirmed by testing for acetylcholine receptor antibodies (seen in 60–80%) or, more rarely, anti-muscle-specific kinase (anti-MuSK) antibodies. Treatment is with the anti-muscarinic drugs neostigmine or pyridostigmine. In the longer term, immunosuppressive therapy with prednisolone or azathioprine is of value. Plasma exchange is used for crises. Thymectomy is considered if a thymoma is present or if the response to medical therapy is unsatisfactory. About a quarter will show remission post thymectomy and up to half show some improvement.



Figure 27.9 Myasthenia gravis showing ptosis from ocular muscle fatigue which improved with edrophonium.

Muscle disorders

The muscular dystrophies

This is a group of inherited disorders with muscle degeneration, often progressive.

Duchenne muscular dystrophy

Duchenne muscular dystrophy is the most common phenotype, affecting 1 in 4000 male infants. It is inherited as an X-linked recessive disorder, although about a third have new mutations. It results from a deletion on the short arm of the X chromosome (at the Xp21 site). This site codes for a protein called dystrophin, which connects the cytoskeleton of a muscle fibre to the surrounding extracellular matrix through the cell membrane. Where it is deficient, there are several aberrant intracellular signalling pathways associated with an influx of calcium ions, a breakdown of the calcium calmodulin complex and an excess of free radicals, ultimately leading to myofibre necrosis. The serum creatine phosphokinase (CPK) is markedly elevated. Some countries, e.g. Wales, have introduced neonatal screening for Duchenne dystrophy; affected children are detected in the neonatal screening test by an elevated CPK.

Children present with a waddling gait and/or language delay; they have to mount stairs one by one and run slowly compared to their peers. Although the average age of diagnosis remains 5.5 years, children often become symptomatic much earlier. They will show Gowers sign (the need to turn prone to rise). There is pseudohypertrophy of the calves because of replacement of muscle fibres by fat and fibrous tissue.

In the early school years, affected boys tend to be slower and clumsier than their peers. The progressive muscle atrophy and weakness means that they are no longer ambulant by the age of about 10–14 years. Life expectancy is reduced to the late twenties from respiratory failure or the associated cardiomyopathy. About one-third of affected children have learning difficulties. Scoliosis is a common complication.

Management – Appropriate exercise helps to maintain muscle power and mobility and delays the onset of scoliosis. Contractures, particularly at the ankles, should be prevented by passive stretching and the provision of night splints. Walking can be prolonged with the provision of orthoses, in particular those which allow ambulation by leaning from side to side. Lengthening of the Achilles tendon may be required to facilitate ambulation. Attention to maintaining a good sitting posture helps to minimise the risk of scoliosis. Scoliosis is managed with a truncal brace, a moulded seat and ultimately surgical insertion of a metal spinal rod. Later in the condition, episodes of nocturnal hypoxia secondary to weakness of the intercostal muscles may present with lassitude or irritability. Respiratory aids, particularly overnight CPAP (continuous

positive airway pressure) or non-invasive positive pressure ventilation (NIPPV), may be provided to improve the quality of life. As with all chronic disabling conditions, parent self-help groups are a useful continuing source of information and support for families. Affected children should be reviewed periodically at a specialist regional centre. Ambulant children with Duchenne dystrophy are increasingly treated with corticosteroids (prednisolone for 10 days each month) to preserve mobility and prevent scoliosis. The precise mechanism by which glucocorticoids may help is not known.

It may be possible to identify female carriers if they have a mildly raised CPK or if the gene deletion can be detected on DNA analysis. Antenatal diagnosis is then possible.

Becker muscular dystrophy

In Becker dystrophy some functional dystrophin is produced. The features are similar to those of Duchenne dystrophy but clinically the disease progresses more slowly. The average age of onset is 11 years, inability to walk in the late twenties, with life expectancy being from the late forties to normal.

Congenital muscular dystrophies

This is a heterogeneous group of disorders, most with recessive inheritance, which present at birth or early infancy with weakness, hypotonia or contractures. Typically the proximal weakness is slowly progressive with a tendency to contracture when the ability to walk is lost. Some may run a more static course. Biopsy shows dystrophic features with a reduction of one of the extracellular matrix proteins such as laminin (most common); or one of several glycosyltransferases. These dystrophies may be linked with central nervous abnormalities, which may result in learning difficulties.

Congenital myopathies

These present at birth or in infancy with generalised hypotonia and muscle weakness. They are static or slowly progressive. They are named according to the changes seen on muscle biopsy or electron microscopy. Creatine phosphokinase levels are normal or only mildly elevated.

Metabolic myopathies

Metabolic conditions can affect muscles, due either to the deposition of storage material or to energy-depleting enzyme deficiencies. Presentation is as a floppy infant or, in older children, with muscle weakness or cramps on exercise. The main causes are:

- Glycogen storage disorders (see Ch. 25).
- Disorders of lipid metabolism. Fatty acids are important muscle fuel. Fatty acid oxidation occurs in the mitochondria and defects in this pathway can result in weakness. Carnitine is essential to supply long-chain fatty acids to the mitochondria for breakdown, and carnitine deficiency causes weakness.
- Mitochondrial cytopathies. Rare disorders which are coded as maternally inherited mitochondrial DNA. Myopathy may be the major manifestation or the disorder may be multisystem, with lactic acidosis and encephalopathy. Mitochondrial DNA testing is available.



Figure 27.10 Heliotrope rash in dermatomyositis.

The inflammatory myopathies

Benign acute myositis

This is assumed to be a post-viral phenomenon, as it often follows an upper respiratory tract infection and runs a self-limiting course. Pain and weakness occur in affected muscles. CPK is usually raised.

Dermatomyositis

This is a systemic illness, probably due to an angiopathy. Usual onset is between 5 and 10 years. This can be acute, but more typically is insidious with fever, misery, and eventually symmetrical muscle weakness, which is mainly proximal. Sometimes pharyngeal muscle involvement affects swallowing. There is also a characteristic violaceous (heliotrope) rash to the eyelids, and periorbital oedema (Fig. 27.10). The rash may also affect the extensor surfaces of joints, e.g. elbow, and with time subcutaneous calcification can appear. Inflammatory markers (CRP, ESR) can be raised but not invariably. Muscle biopsy shows an inflammatory cell infiltrate and atrophy. Physiotherapy is needed to prevent contractures. Corticosteroids are the standard treatment, and continue at a tailored dose for 2 years. Other immunosuppressants, e.g. methotrexate, ciclosporin, may be needed. Mortality is 5–10%.

Myotonic disorders

Myotonia is delayed relaxation after sustained muscle contraction. It can be identified clinically and on electromyography.

Dystrophia myotonica

This relatively common illness is dominantly inherited and caused by a nucleotide triplet repeat expansion, so this means there can be anticipation through generations, especially when maternally transmitted (see Ch. 8). Newborns can present with hypotonia and feeding and respiratory difficulties due to muscle weakness. It is then useful to examine the mother for myotonia. This manifests as slow release of handshake or difficulty releasing the tightly clasped fist. This may be mild and not have been appreciated. Sensitivity is required as diagnosis in the neonate may have repercussions for the family. Older children can present with myopathic facies (Fig. 27.11), learning difficulties and myotonia. Adults develop cataracts and males develop baldness and testicular atrophy. Death is usually due to cardiomyopathy.



Figure 27.11 Dystrophia myotonica in an 8-year-old who has marked facial weakness and moderately severe learning difficulties.

The 'floppy infant'

Persisting hypotonia in infants can be readily felt on picking up the infant, who tends to slip through the fingers or hang like a rag doll when suspended prone. There will be marked head lag when the head is lifted by the arms from supine. The causes are listed in Box 27.3. The clinical examination may help determine the site of the lesion, whether cortical or neuromuscular. Central hypotonia is associated with poor truncal tone but preserved limb tone. Dysmorphic features suggest a genetic cause. Lower motor neurone lesions are suggested by a frog-like posture (Fig. 27.7), poor antigravity movements and absent reflexes.

Summary

Neuromuscular disorders

- Present with muscle weakness, which may manifest with floppiness, delayed motor milestones, unsteady gait or muscle fatiguability
- Anterior horn cell – *spinal muscular atrophy*: progressive weakness and wasting of skeletal muscles; tongue fasciculation may aid diagnosis
- Peripheral nerve
  - *Hereditary motor sensory neuropathies* (HMSN): symmetrical wasting of the distal muscles
  - *Acute post-infectious polyneuropathy* (Guillain–Barré syndrome): ascending symmetrical weakness; may be bulbar palsy and respiratory depression
- Neuromuscular transmission – *juvenile myasthenia*: >10 years old, ophthalmoplegia and ptosis, loss of facial expression and difficulty chewing
- Muscle – *Duchenne muscular dystrophy*: X-linked recessive, presents with waddling gait and difficulty climbing stairs

Box 27.3 Causes of the floppy (hypotonic) infant

Central

Cortical

- Hypoxic-ischaemic encephalopathy
- Cortical malformations

Genetic

- Down syndrome
- Prader–Willi syndrome

Metabolic

- Hypothyroidism
- Hypocalcaemia

Peripheral

Neuromuscular

- Spinal muscular atrophy
- Myopathy
- Myotonia
- Congenital myasthenia.

Ataxia

Friedreich ataxia

This is an autosomal recessive condition. The gene mutation (Frataxin) is an example of a trinucleotide repeat disorder. It presents with worsening ataxia, distal wasting in the legs, absent lower limb reflexes but extensor plantar responses because of pyramidal involvement, pes cavus and dysarthria. This is similar to the hereditary motor sensory neuropathies, but in Friedreich ataxia, there is impairment of joint position and vibration sense, extensor plantars and there is often optic atrophy. The cerebellar component becomes more apparent with age. Evolving kyphoscoliosis and cardiomyopathy can cause cardiorespiratory compromise and death at 40–50 years.

Ataxia telangiectasia

This disorder of DNA repair is an autosomal recessive condition. The gene (ATM) has been identified. There may be mild delay in motor development in infancy and oculomotor problems with incoordination and delay in ocular pursuit of objects (oculomotor dyspraxia), with difficulty with balance and coordination becoming evident at school age. There is subsequent deterioration, with a mixture of dystonia and cerebellar signs. Many children require a wheelchair for mobility in early adolescence. Telangiectasia develops in the conjunctiva (Fig. 27.12), neck and shoulders from about 4 years of age. These children:

- Have an increased susceptibility to infection, principally from an IgA surface antibody defect
- Develop malignant disorders, principally acute lymphoblastic leukaemia (about 10%)
- Have a raised serum alpha-fetoprotein



Figure 27.12 Telangiectasia of the conjunctiva are present from about 4 years of age in ataxia telangiectasia.

- Have an increased white cell sensitivity to irradiation, which can be used diagnostically, but the ATM gene test is now mostly used.

Other hereditary cerebellar ataxias

There is a growing number of these, largely dominantly inherited (genotypes identified), with a relatively benign course in childhood.

Summary

Cerebellar ataxia

- Causes – medication and drugs, varicella infection, posterior fossa lesions or tumours, genetic and degenerative disorders, e.g. ataxic cerebral palsy, Friedreich ataxia and ataxia telangiectasia.

Cerebrovascular disease

Intracranial haemorrhage

Extradural haemorrhage

This usually follows direct head trauma, often associated with skull fracture (tearing of middle meningeal artery as it passes through the foramen spinosum of the sphenoid bone). It results from arterial or venous bleeding into the extradural space. There is often a lucid interval until the conscious level deteriorates, with seizures, secondary to increasing size of the haematoma. There may be focal neurological signs with dilatation of the ipsilateral pupil, paresis of the contralateral limbs and a false localising uni- or bilateral VIth nerve paresis. In young children, initial presentation may be with anaemia and shock. The diagnosis is confirmed with a CT scan. Management is to correct hypovolaemia, urgent evacuation of the haematoma and arrest of the bleeding.

Subdural haematoma

This results from tearing of the veins as they cross the subdural space. It is a characteristic lesion in non-accidental injury caused by shaking or direct trauma in infants or toddlers. Retinal haemorrhages are usually present. There has been recent controversy surrounding the relative contributions of direct trauma, shearing injury and hypoxia. Subdural haematomas are occasionally seen following a fall from a considerable height.

Subarachnoid haemorrhage

This is much more common in adults. Presentation is usually with acute onset of head pain, neck stiffness and occasionally fever. Retinal haemorrhage is usually present. Seizures and coma may develop. A CT scan of the head usually identifies blood in the CSF. A lumbar puncture in the acute situation is best avoided as haemorrhage may extend following the release of intracranial pressure. The cause is often an aneurysm or arteriovenous malformation (AVM). It can be identified on MR angiography (MRA), CT or conventional angiography. Treatment can be neurosurgical or with interventional radiography.

Summary

Cerebral haemorrhage

- History of significant head injury – remember that an extradural haemorrhage may be present even if lucid afterwards
- Subdural haematoma and retinal haemorrhages in an infant – consider non-accidental injury caused by shaking or direct trauma.

Stroke

Infantile stroke is dealt with in Chapter 10. Childhood stroke may be due to vascular, thromboembolic or haemorrhagic disease. The clinical presentation is determined by the vascular territory involved. There is compromise of the anterior circulation (internal carotid, anterior and middle cerebral arteries), which leads to hemiparesis with or without speech disturbance. Less common is compromise of the posterior circulation (vertebrobasilar arteries) with associated visual or cerebellar signs.

Causes include:

- Cardiac: congenital cyanotic heart disease, e.g. Fallot tetralogy, endocarditis
- Haematological: sickle cell disease; deficiencies of anti-thrombotic factors, e.g. protein S
- Post-infective: following varicella or other viral infection
- Inflammatory: damage to vessels in autoimmune disease, e.g. SLE (systemic lupus erythematosus)
- Metabolic/genetic: homocystinuria, mitochondrial disorders, e.g. MELAS (**m**yoclonic **e**pilepsy, **l**actic **a**cidosis and **s**troke); CADASIL (**c**erebral **a**utosomal **d**ominant **a**rteriopathy with **s**ubcortical **i**nfarcts